THE 180-DAY SOFTWARE ENGINEERING INTERVIEW PREPARATION HANDBOOK

Java DSA • CS Fundamentals • System Design • Backend Project • Behavioral • Summer 2027 Recruiting

Program dates: Wednesday, October 14, 2026 (Day 1) through Sunday, April 11, 2027 (Day 180)

Candidate profile: Junior Computer Science student at the University of North Texas (UNT), F-1 student visa

Minimum commitment: 3 focused hours every day • 21 hours per week • 540 hours over 180 days

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Designed as a practical training manual—not an offer guarantee. Follow the daily plan, measure independent performance, and adjust the workload based on evidence.

Version date: October 13, 2026

Table of Contents

1. **How to use this handbook** — Starting assumptions, readiness criteria, diagnostic and tracking
2. **Six-month overview** — Monthly outcomes and weekly cadence
3. **Spaced-repetition system** — R0/R1/R3/R7/R14/R30/R60 rules and review ledger

5. Backend project plan · Beginner-friendly Campus Task Tracker API

6. **Behavioral question bank** — Authentic STAR practice throughout the program
7. **Recruiting and F-1/CPT checklist** — Summer 2027 recruiting actions and official guidance
8. **Daily tracker and readiness checklist** — Reusable log, mastery scale, final criteria
9. **Problem index and references** — Problem links and official sources

1. How to Use This Handbook

This plan assumes you know basic Java syntax and can write small programs, but your DSA speed, pattern recognition, and interview communication still need structured practice. Java is the primary DSA language. Your portfolio track is a beginner-friendly Campus Task Tracker API: start with a small Java console app, then add simple Spring Boot endpoints, persistence, and tests as your knowledge grows. Build each part yourself and only include features you can explain. Keep the project bounded so it does not displace DSA.

If you are a beginner: use the first 30 days to slow down, solve fewer problems, and implement each pattern twice. Keep the 3-hour minimum, but use more time for guided examples and tests. If intermediate: attempt the problem cold first, then use hints only after a genuine attempt; use the optional stretch problems and stricter timers.

Daily three-hour rule

Every daily plan totals 180 minutes. On holidays, keep the commitment but use a lighter format: active recall, re-solves, a short mock, project notes, or recruiting maintenance. A review is successful only when you recall or solve independently—not when you reread a solution.

Mastery scale

Status	Definition
Not Tried	No genuine attempt yet.
Tried	Attempted, but needed substantial hints or could not finish.
Developing	Can explain part of the approach; still misses edge cases or implementation details.
Strong	Solves independently within a reasonable interview time, explains correctness and complexity, and tests edge cases.
Mastered	Repeated independent success on spaced reviews, including a cold re-solve and a clear explanation; not merely solution recognition.

Interview readiness indicators

• DSA: independently solve a representative medium in roughly 25–35 minutes, with clear narration and tested edge cases; performance should be repeatable across multiple patterns.

• Communication: ask clarifying questions, describe a baseline, derive an optimization, justify correctness, analyze time/space, and test.

• CS/design: explain core OOP, OS, SQL/database, networking, and basic system-design trade-offs without relying on memorized slogans.

• Project: run the project from documented instructions, explain architecture and failure cases, demonstrate tests, and describe honest limitations.

• Behavioral/recruiting: deliver concise, authentic STAR stories and maintain a verified application tracker.

First-three-day diagnostic

Day 1: Two Sum and Valid Anagram, with a cold attempt and Big-O explanation. Day 2: implement array reverse and binary search; test boundaries and duplicates. Day 3: Valid Parentheses, Java collections review, and a 2-minute project walkthrough. Record solve time, hints, bugs, and confidence (1–5). Do not use this as a pass/fail judgment; it is your baseline.

Weak-topic allocation rule

At each weekly checkpoint, classify misses as (1) pattern recognition, (2) algorithm reasoning, (3) Java implementation, (4) edge-case testing, or (5) communication. Give the weakest category an extra 20 minutes on three non-assessment days next week, replacing optional stretch work—not the spaced-review block. If a review fails, add a short next-day reattempt and keep the original R dates in the ledger.

2. Six-Month Overview

Period	Focus	Milestone
Days 1–30 Oct 14–Nov 12	Java foundations and essential DSA	Month 1 assessment; clean Java templates; error log started.
Days 31–60 Nov 13–Dec 12	Core interview patterns	Trees, backtracking, heaps, tries; first mock round.
Days 61–90 Dec 13–Jan 11	Advanced algorithms	Graph and DP patterns; timed mixed sets.
Days 91–120 Jan 12–Feb 10	CS fundamentals and system design	OOP/LLD, OS, SQL, networking, basic system design.
Days 121–150 Feb 11–Mar 12	Project, mixed practice, recruiting	Working backend project slice, tests, README, resume, STAR bank.
Days 151–180 Mar 13–Apr 11	Interview simulation and final prep	Mocks, weak-area repair, readiness assessment and next steps.

Weekly rhythm

Days 1–6 of each week combine new learning, spaced review, practical application, and verbal explanation. Every seventh day is a 3-hour assessment/review day. Every fourth week includes a cumulative checkpoint. Monthly assessments occur on Days 30, 60, 90, 120, 150, and 180. Since some checkpoint days coincide, the cumulative/monthly assessment replaces the ordinary weekly test rather than adding extra hours.

Weekly objective index

Week 1 (Days 1–7): Baseline diagnostic, Java fluency, arrays and hashing.

Week 2 (Days 8–14): Two pointers, sliding windows, prefix sums.

Week 3 (Days 15–21): Sorting, binary search, recursion.

Week 4 (Days 22–28): Linked lists, stacks/queues, Month 1 checkpoint.

Week 5 (Days 29–35): Fast/slow pointers, intervals, monotonic structures.

Week 6 (Days 36–42): Trees, backtracking, heaps and tries.

Week 7 (Days 43–49): Greedy, matrices, timed core-pattern practice.

Week 8 (Days 50–56): Mixed core patterns and Month 2 checkpoint.

Week 9 (Days 57–63): Graph representations, DFS/BFS, cycle detection.

Week 10 (Days 64–70): Topological sort, Union-Find, shortest paths.

Week 11 (Days 71–77): DP state design, 1D/2D DP, knapsack.

Week 12 (Days 78–84): Subsequences, bit manipulation, mixed advanced DSA.

Week 13 (Days 85–91): OOP/LLD, OS, databases, networking.

Week 14 (Days 92–98): System design fundamentals and Month 4 checkpoint.

Week 15 (Days 99–105): Build backend API and database integration.

Week 16 (Days 106–112): Testing, Docker, CI/CD, observability.

Week 17 (Days 113–119): Project narrative, resume, STAR stories, recruiting.

Week 18 (Days 120–126): Mixed practice, OA simulation, Month 5 checkpoint.

Week 19 (Days 127–133): Timed DSA rounds and graph/DP refresh.

Week 20 (Days 134–140): Mock technical interviews and weak-pattern repair.

Week 21 (Days 141–147): Behavioral, system design and project deep dives.

Week 22 (Days 148–154): Company-style assessments and final revisions.

Week 23 (Days 155–161): Final readiness, confidence, sleep and execution.

Week 24 (Days 162–168): Consolidate prior patterns; test independently; repair the weakest skill.

Week 25 (Days 169–175): Consolidate prior patterns; test independently; repair the weakest skill.

Week 26 (Days 176–180): Consolidate prior patterns; test independently; repair the weakest skill.

3. Spaced-Repetition System

For each day's anchor topic and anchor problem, R0 is the original learning day. The review dates are calculated from that original date—not from the previous review: R1 = +1 day; R3 = +3; R7 = +7; R14 = +14; R30 = +30; R60 = +60. For example, Day 1 (Oct 14) reviews on Oct 15, Oct 17, Oct 21, Oct 28, Nov 13, and Dec 13. The same offset rules apply to every daily anchor, and the daily entries show the exact due queue.

Review protocol (5–7 minutes per anchor, batch related concepts)

• Recall the concept without notes; say the trigger conditions and invariant.

• Re-solve the anchor problem from a blank editor or explain a correct implementation line by line.

• State time and space complexity, then test at least two edge cases.

• Identify the prior mistake and what prevents it from recurring.

• Mark Strong, Needs Practice, or Relearn. If it fails, add a next-day repair attempt; do not erase the original review dates.

Workload control

Each day has one anchor topic/problem. Due anchors are grouped into the 30–35 minute review block; prioritize Relearn and Needs Practice first, and use short verbal recall for Strong items. The full due queue is listed in the daily entry. If the queue exceeds the block, complete the weakest items fully and use rapid closed-book recall for strong items, then continue the remaining queue in the next available review block. Never mark a problem successful just because you read the solution.

Calendar examples

Day 1 — Wednesday, October 14, 2026: R1: Day 2 (Oct 15, 2026); R3: Day 4 (Oct 17, 2026); R7: Day 8 (Oct 21, 2026); R14: Day 15 (Oct 28, 2026); R30: Day 31 (Nov 13, 2026); R60: Day 61 (Dec 13, 2026)

Day 10 — Friday, October 23, 2026: R1: Day 11 (Oct 24, 2026); R3: Day 13 (Oct 26, 2026); R7: Day 17 (Oct 30, 2026); R14: Day 24 (Nov 06, 2026); R30: Day 40 (Nov 22, 2026); R60: Day 70 (Dec 22, 2026)

Day 30 — Thursday, November 12, 2026: R1: Day 31 (Nov 13, 2026); R3: Day 33 (Nov 15, 2026); R7: Day 37 (Nov 19, 2026); R14: Day 44 (Nov 26, 2026); R30: Day 60 (Dec 12, 2026); R60: Day 90 (Jan 11, 2027)

Day 60 — Saturday, December 12, 2026: R1: Day 61 (Dec 13, 2026); R3: Day 63 (Dec 15, 2026); R7: Day 67 (Dec 19, 2026); R14: Day 74 (Dec 26, 2026); R30: Day 90 (Jan 11, 2027); R60: Day 120 (Feb 10, 2027)

Day 90 — Monday, January 11, 2027: R1: Day 91 (Jan 12, 2027); R3: Day 93 (Jan 14, 2027); R7: Day 97 (Jan 18, 2027); R14: Day 104 (Jan 25, 2027); R30: Day 120 (Feb 10, 2027); R60: Day 150 (Mar 12, 2027)

Day 120 — Wednesday, February 10, 2027: R1: Day 121 (Feb 11, 2027); R3: Day 123 (Feb 13, 2027); R7: Day 127 (Feb 17, 2027); R14: Day 134 (Feb 24, 2027); R30: Day 150 (Mar 12, 2027); R60: Day 180 (Apr 11, 2027)

4. Complete 180-Day Daily Schedule

Every entry below is dated and includes at least 180 minutes of focused study. Problem difficulty is from LeetCode's displayed classification and may change; always use the linked problem page. When a problem is assigned as an anchor, make a genuine attempt before reading editorials. On a day centered on CS/project work, the anchor is still a short DSA problem so the spaced-repetition system remains consistent.

Day 1 — Week 1 — Wednesday, October 14, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Java collections, generics, equals/hashCode

Learning objective: Understand the key invariant or decision rule for java collections, generics, equals/hashcode; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Longest Consecutive Sequence — Medium; pattern: HashSet sequence starts. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Diagnostic / retrieval: Write what you know before looking anything up; mark confidence.

45 min — Focused lesson: Learn java collections, generics, equals/hashcode; implement a tiny Java example and state the invariant.

60 min — Problem attempt: Attempt the listed problem with a timer; no solution for the first 25 minutes.

30 min — CS/project/recruiting: Complete the specific secondary task for this day.

15 min — Error log + recall: Record complexity, mistake, and review status.

Spaced-repetition queue due today: None yet; use the review block for same-day recall and diagnostic notes.

Secondary task: Baseline: solve Two Sum and Valid Anagram; note where you got stuck.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Longest Consecutive Sequence* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 2 (Oct 15), R3 Day 4 (Oct 17), R7 Day 8 (Oct 21), R14 Day 15 (Oct 28), R30 Day 31 (Nov 13), R60 Day 61 (Dec 13)

Day 2 — Week 1 — Thursday, October 15, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Big-O, arrays, loop invariants

Learning objective: Understand the key invariant or decision rule for big-o, arrays, loop invariants; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Top K Frequent Elements — Medium; pattern: frequency map + heap/buckets. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Diagnostic / retrieval: Write what you know before looking anything up; mark confidence.

45 min — Focused lesson: Learn big-o, arrays, loop invariants; implement a tiny Java example and state the invariant.

60 min — Problem attempt: Attempt the listed problem with a timer; no solution for the first 25 minutes.

30 min — CS/project/recruiting: Complete the specific secondary task for this day.

15 min — Error log + recall: Record complexity, mistake, and review status.

Spaced-repetition queue due today: Day 1 R1: Longest Consecutive Sequence

Secondary task: Baseline: implement array reverse and binary search; explain Big-O.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Top K Frequent Elements* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 3 (Oct 16), R3 Day 5 (Oct 18), R7 Day 9 (Oct 22), R14 Day 16 (Oct 29), R30 Day 32 (Nov 14), R60 Day 62 (Dec 14)

Day 3 — Week 1 — Friday, October 16, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Strings, StringBuilder, character counts

Learning objective: Understand the key invariant or decision rule for strings, stringbuilder, character counts; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Encode and Decode Strings — Medium; pattern: serialization boundaries. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Diagnostic / retrieval: Write what you know before looking anything up; mark confidence.

45 min — Focused lesson: Learn strings, stringbuilder, character counts; implement a tiny Java example and state the invariant.

60 min — Problem attempt: Attempt the listed problem with a timer; no solution for the first 25 minutes.

30 min — CS/project/recruiting: Complete the specific secondary task for this day.

15 min — Error log + recall: Record complexity, mistake, and review status.

Spaced-repetition queue due today: Day 2 R1: Top K Frequent Elements

Secondary task: Baseline: solve Valid Parentheses; explain Java collections and outline one project.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Encode and Decode Strings* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 4 (Oct 17), R3 Day 6 (Oct 19), R7 Day 10 (Oct 23), R14 Day 17 (Oct 30), R30 Day 33 (Nov 15), R60 Day 63 (Dec 15)

Day 4 — Week 1 — Saturday, October 17, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: HashMap/HashSet patterns

Learning objective: Understand the key invariant or decision rule for hashmap/hashset patterns; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Product of Array Except Self — Medium; pattern: prefix/suffix products. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study hashmap/hashset patterns; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Product of Array Except Self cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 1 R3: Longest Consecutive Sequence; Day 3 R1: Encode and Decode Strings

Secondary task: Review Java pitfalls: equals/hashCode, nulls, String immutability, integer overflow.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Product of Array Except Self* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 5 (Oct 18), R3 Day 7 (Oct 20), R7 Day 11 (Oct 24), R14 Day 18 (Oct 31), R30 Day 34 (Nov 16), R60 Day 64 (Dec 16)

Day 5 — Week 1 — Sunday, October 18, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Two Sum and complement reasoning

Learning objective: Understand the key invariant or decision rule for two sum and complement reasoning; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study two sum and complement reasoning; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Single Number cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 2 R3: Top K Frequent Elements; Day 4 R1: Product of Array Except Self

Secondary task: Add one mistake and one corrected test case to the error log.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Single Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 6 (Oct 19), R3 Day 8 (Oct 21), R7 Day 12 (Oct 25), R14 Day 19 (Nov 01), R30 Day 35 (Nov 17), R60 Day 65 (Dec 17)

Day 6 — Week 1 — Monday, October 19, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Two pointers and palindrome checks

Learning objective: Understand the key invariant or decision rule for two pointers and palindrome checks; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Valid Parentheses — Easy; pattern: stack. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study two pointers and palindrome checks; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Valid Parentheses cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 3 R3: Encode and Decode Strings; Day 5 R1: Single Number

Secondary task: Implement a tiny Java example using ArrayList, HashMap, HashSet, and generics.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Valid Parentheses* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 7 (Oct 20), R3 Day 9 (Oct 22), R7 Day 13 (Oct 26), R14 Day 20 (Nov 02), R30 Day 36 (Nov 18), R60 Day 66 (Dec 18)

Day 7 — Week 1 — Tuesday, October 20, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: Java + arrays

Learning objective: Understand the key invariant or decision rule for weekly assessment: java + arrays; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Product of Array Except Self — Medium; pattern: prefix/suffix products. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering baseline diagnostic, java fluency, arrays and hashing; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Two pointers, sliding windows, prefix sums.

Spaced-repetition queue due today: Day 4 R3: Product of Array Except Self; Day 6 R1: Valid Parentheses

Secondary task: Write brute-force and optimized approaches; explain Big-O for both.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 8 (Oct 21), R3 Day 10 (Oct 23), R7 Day 14 (Oct 27), R14 Day 21 (Nov 03), R30 Day 37 (Nov 19), R60 Day 67 (Dec 19)

Week 1 checkpoint: Baseline diagnostic, Java fluency, arrays and hashing. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 8 — Week 2 — Wednesday, October 21, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Frequency maps and anagram grouping

Learning objective: Understand the key invariant or decision rule for frequency maps and anagram grouping; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Group Anagrams — Medium; pattern: canonical key / hashing. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study frequency maps and anagram grouping; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Group Anagrams cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 1 R7: Longest Consecutive Sequence; Day 5 R3: Single Number; Day 7 R1: Product of Array Except Self

Secondary task: Create edge-case tests: empty input, one element, duplicates, negatives, overflow.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Group Anagrams* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 9 (Oct 22), R3 Day 11 (Oct 24), R7 Day 15 (Oct 28), R14 Day 22 (Nov 04), R30 Day 38 (Nov 20), R60 Day 68 (Dec 20)

Day 9 — Week 2 — Thursday, October 22, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Prefix and suffix sums/products

Learning objective: Understand the key invariant or decision rule for prefix and suffix sums/products; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Longest Consecutive Sequence — Medium; pattern: HashSet sequence starts. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study prefix and suffix sums/products; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Longest Consecutive Sequence cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 2 R7: Top K Frequent Elements; Day 6 R3: Valid Parentheses; Day 8 R1: Group Anagrams

Secondary task: Explain the pattern aloud in 3 minutes as if to an interviewer.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Longest Consecutive Sequence* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 10 (Oct 23), R3 Day 12 (Oct 25), R7 Day 16 (Oct 29), R14 Day 23 (Nov 05), R30 Day 39 (Nov 21), R60 Day 69 (Dec 21)

Day 10 — Week 2 — Friday, October 23, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Sliding window: fixed-size windows

Learning objective: Understand the key invariant or decision rule for sliding window: fixed-size windows; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: 3Sum — Medium; pattern: sort + two pointers. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study sliding window: fixed-size windows; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt 3Sum cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 3 R7: Encode and Decode Strings; Day 7 R3: Product of Array Except Self; Day 9 R1: Longest Consecutive Sequence

Secondary task: Review Java pitfalls: equals/hashCode, nulls, String immutability, integer overflow.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *3Sum* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 11 (Oct 24), R3 Day 13 (Oct 26), R7 Day 17 (Oct 30), R14 Day 24 (Nov 06), R30 Day 40 (Nov 22), R60 Day 70 (Dec 22)

Day 11 — Week 2 — Saturday, October 24, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Sliding window: variable-size windows

Learning objective: Understand the key invariant or decision rule for sliding window: variable-size windows; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Minimum Window Substring — Hard; pattern: window counts. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study sliding window: variable-size windows; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Minimum Window Substring cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 4 R7: Product of Array Except Self; Day 8 R3: Group Anagrams; Day 10 R1: 3Sum

Secondary task: Add one mistake and one corrected test case to the error log.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Minimum Window Substring* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 12 (Oct 25), R3 Day 14 (Oct 27), R7 Day 18 (Oct 31), R14 Day 25 (Nov 07), R30 Day 41 (Nov 23), R60 Day 71 (Dec 23)

Day 12 — Week 2 — Sunday, October 25, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Sorting, comparators, stable ordering

Learning objective: Understand the key invariant or decision rule for sorting, comparators, stable ordering; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find Minimum in Rotated Sorted Array — Medium; pattern: partition invariant. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study sorting, comparators, stable ordering; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Find Minimum in Rotated Sorted Array cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 5 R7: Single Number; Day 9 R3: Longest Consecutive Sequence; Day 11 R1: Minimum Window Substring

Secondary task: Implement a tiny Java example using ArrayList, HashMap, HashSet, and generics.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Find Minimum in Rotated Sorted Array* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 13 (Oct 26), R3 Day 15 (Oct 28), R7 Day 19 (Nov 01), R14 Day 26 (Nov 08), R30 Day 42 (Nov 24), R60 Day 72 (Dec 24)

Day 13 — Week 2 — Monday, October 26, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Binary search invariants

Learning objective: Understand the key invariant or decision rule for binary search invariants; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Binary Search — Easy; pattern: search invariant. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study binary search invariants; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Binary Search cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 6 R7: Valid Parentheses; Day 10 R3: 3Sum; Day 12 R1: Find Minimum in Rotated Sorted Array

Secondary task: Write brute-force and optimized approaches; explain Big-O for both.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Binary Search* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 14 (Oct 27), R3 Day 16 (Oct 29), R7 Day 20 (Nov 02), R14 Day 27 (Nov 09), R30 Day 43 (Nov 25), R60 Day 73 (Dec 25)

Day 14 — Week 2 — Tuesday, October 27, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Binary search edge cases and answer-space search

Learning objective: Understand the key invariant or decision rule for binary search edge cases and answer-space search; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Merge Intervals — Medium; pattern: sort + sweep. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering two pointers, sliding windows, prefix sums; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Sorting, binary search, recursion.

Spaced-repetition queue due today: Day 7 R7: Product of Array Except Self; Day 11 R3: Minimum Window Substring; Day 13 R1: Binary Search

Secondary task: Create edge-case tests: empty input, one element, duplicates, negatives, overflow.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 15 (Oct 28), R3 Day 17 (Oct 30), R7 Day 21 (Nov 03), R14 Day 28 (Nov 10), R30 Day 44 (Nov 26), R60 Day 74 (Dec 26)

Week 2 checkpoint: Two pointers, sliding windows, prefix sums. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 15 — Week 3 — Wednesday, October 28, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: arrays + search

Learning objective: Understand the key invariant or decision rule for weekly assessment: arrays + search; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Product of Array Except Self — Medium; pattern: prefix/suffix products. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study weekly assessment: arrays + search; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Product of Array Except Self cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 1 R14: Longest Consecutive Sequence; Day 8 R7: Group Anagrams; Day 12 R3: Find Minimum in Rotated Sorted Array; Day 14 R1: Merge Intervals

Secondary task: Explain the pattern aloud in 3 minutes as if to an interviewer.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Product of Array Except Self* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 16 (Oct 29), R3 Day 18 (Oct 31), R7 Day 22 (Nov 04), R14 Day 29 (Nov 11), R30 Day 45 (Nov 27), R60 Day 75 (Dec 27)

Day 16 — Week 3 — Thursday, October 29, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Recursion, base cases, call stack

Learning objective: Understand the key invariant or decision rule for recursion, base cases, call stack; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Reverse Linked List — Easy; pattern: pointer rewiring. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study recursion, base cases, call stack; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Reverse Linked List cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 2 R14: Top K Frequent Elements; Day 9 R7: Longest Consecutive Sequence; Day 13 R3: Binary Search; Day 15 R1: Product of Array Except Self

Secondary task: Review Java pitfalls: equals/hashCode, nulls, String immutability, integer overflow.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Reverse Linked List* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 17 (Oct 30), R3 Day 19 (Nov 01), R7 Day 23 (Nov 05), R14 Day 30 (Nov 12), R30 Day 46 (Nov 28), R60 Day 76 (Dec 28)

Day 17 — Week 3 — Friday, October 30, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Linked list pointer manipulation

Learning objective: Understand the key invariant or decision rule for linked list pointer manipulation; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Unique Paths — Medium; pattern: grid DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study linked list pointer manipulation; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Unique Paths cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 3 R14: Encode and Decode Strings; Day 10 R7: 3Sum; Day 14 R3: Merge Intervals; Day 16 R1: Reverse Linked List

Secondary task: Add one mistake and one corrected test case to the error log.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Unique Paths* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 18 (Oct 31), R3 Day 20 (Nov 02), R7 Day 24 (Nov 06), R14 Day 31 (Nov 13), R30 Day 47 (Nov 29), R60 Day 77 (Dec 29)

Day 18 — Week 3 — Saturday, October 31, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Fast/slow pointers

Learning objective: Understand the key invariant or decision rule for fast/slow pointers; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Linked List Cycle — Easy; pattern: fast/slow pointers. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study fast/slow pointers; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Linked List Cycle cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 4 R14: Product of Array Except Self; Day 11 R7: Minimum Window Substring; Day 15 R3: Product of Array Except Self; Day 17 R1: Unique Paths

Secondary task: Implement a tiny Java example using ArrayList, HashMap, HashSet, and generics.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Linked List Cycle* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 19 (Nov 01), R3 Day 21 (Nov 03), R7 Day 25 (Nov 07), R14 Day 32 (Nov 14), R30 Day 48 (Nov 30), R60 Day 78 (Dec 30)

Day 19 — Week 3 — Sunday, November 01, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Stacks with ArrayDeque

Learning objective: Understand the key invariant or decision rule for stacks with arraydeque; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Daily Temperatures — Medium; pattern: monotonic stack. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study stacks with arraydeque; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Daily Temperatures cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 5 R14: Single Number; Day 12 R7: Find Minimum in Rotated Sorted Array; Day 16 R3: Reverse Linked List; Day 18 R1: Linked List Cycle

Secondary task: Write brute-force and optimized approaches; explain Big-O for both.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Daily Temperatures* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 20 (Nov 02), R3 Day 22 (Nov 04), R7 Day 26 (Nov 08), R14 Day 33 (Nov 15), R30 Day 49 (Dec 01), R60 Day 79 (Dec 31)

Day 20 — Week 3 — Monday, November 02, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Queues and BFS intuition

Learning objective: Understand the key invariant or decision rule for queues and bfs intuition; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Merge Two Sorted Lists — Easy; pattern: dummy node. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study queues and bfs intuition; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Merge Two Sorted Lists cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 6 R14: Valid Parentheses; Day 13 R7: Binary Search; Day 17 R3: Unique Paths; Day 19 R1: Daily Temperatures

Secondary task: Create edge-case tests: empty input, one element, duplicates, negatives, overflow.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Merge Two Sorted Lists* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 21 (Nov 03), R3 Day 23 (Nov 05), R7 Day 27 (Nov 09), R14 Day 34 (Nov 16), R30 Day 50 (Dec 02), R60 Day 80 (Jan 01)

Day 21 — Week 3 — Tuesday, November 03, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Monotonic stack introduction

Learning objective: Understand the key invariant or decision rule for monotonic stack introduction; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Linked List Cycle — Easy; pattern: fast/slow pointers. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering sorting, binary search, recursion; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Linked lists, stacks/queues, Month 1 checkpoint.

Spaced-repetition queue due today: Day 7 R14: Product of Array Except Self; Day 14 R7: Merge Intervals; Day 18 R3: Linked List Cycle; Day 20 R1: Merge Two Sorted Lists

Secondary task: Explain the pattern aloud in 3 minutes as if to an interviewer.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 22 (Nov 04), R3 Day 24 (Nov 06), R7 Day 28 (Nov 10), R14 Day 35 (Nov 17), R30 Day 51 (Dec 03), R60 Day 81 (Jan 02)

Week 3 checkpoint: Sorting, binary search, recursion. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 22 — Week 4 — Wednesday, November 04, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: linked lists + stacks

Learning objective: Understand the key invariant or decision rule for weekly assessment: linked lists + stacks; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Unique Paths — Medium; pattern: grid DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study weekly assessment: linked lists + stacks; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Unique Paths cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 8 R14: Group Anagrams; Day 15 R7: Product of Array Except Self; Day 19 R3: Daily Temperatures; Day 21 R1: Linked List Cycle

Secondary task: Review Java pitfalls: equals/hashCode, nulls, String immutability, integer overflow.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Unique Paths* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 23 (Nov 05), R3 Day 25 (Nov 07), R7 Day 29 (Nov 11), R14 Day 36 (Nov 18), R30 Day 52 (Dec 04), R60 Day 82 (Jan 03)

Day 23 — Week 4 — Thursday, November 05, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Java null safety and integer overflow

Learning objective: Understand the key invariant or decision rule for java null safety and integer overflow; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Two Sum — Easy; pattern: HashMap complement. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study java null safety and integer overflow; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Two Sum cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 9 R14: Longest Consecutive Sequence; Day 16 R7: Reverse Linked List; Day 20 R3: Merge Two Sorted Lists; Day 22 R1: Unique Paths

Secondary task: Add one mistake and one corrected test case to the error log.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Two Sum* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 24 (Nov 06), R3 Day 26 (Nov 08), R7 Day 30 (Nov 12), R14 Day 37 (Nov 19), R30 Day 53 (Dec 05), R60 Day 83 (Jan 04)

Day 24 — Week 4 — Friday, November 06, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Matrix traversal and boundary handling

Learning objective: Understand the key invariant or decision rule for matrix traversal and boundary handling; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Binary Search — Easy; pattern: search invariant. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study matrix traversal and boundary handling; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Binary Search cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 10 R14: 3Sum; Day 17 R7: Unique Paths; Day 21 R3: Linked List Cycle; Day 23 R1: Two Sum

Secondary task: Implement a tiny Java example using ArrayList, HashMap, HashSet, and generics.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Binary Search* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 25 (Nov 07), R3 Day 27 (Nov 09), R7 Day 31 (Nov 13), R14 Day 38 (Nov 20), R30 Day 54 (Dec 06), R60 Day 84 (Jan 05)

Day 25 — Week 4 — Saturday, November 07, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Prefix sums and subarray reasoning

Learning objective: Understand the key invariant or decision rule for prefix sums and subarray reasoning; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Longest Consecutive Sequence — Medium; pattern: HashSet sequence starts. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study prefix sums and subarray reasoning; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Longest Consecutive Sequence cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 11 R14: Minimum Window Substring; Day 18 R7: Linked List Cycle; Day 22 R3: Unique Paths; Day 24 R1: Binary Search

Secondary task: Write brute-force and optimized approaches; explain Big-O for both.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Longest Consecutive Sequence* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 26 (Nov 08), R3 Day 28 (Nov 10), R7 Day 32 (Nov 14), R14 Day 39 (Nov 21), R30 Day 55 (Dec 07), R60 Day 85 (Jan 06)

Day 26 — Week 4 — Sunday, November 08, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Merge intervals and sort-sweep

Learning objective: Understand the key invariant or decision rule for merge intervals and sort-sweep; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search in Rotated Sorted Array — Medium; pattern: modified binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study merge intervals and sort-sweep; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Search in Rotated Sorted Array cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 12 R14: Find Minimum in Rotated Sorted Array; Day 19 R7: Daily Temperatures; Day 23 R3: Two Sum; Day 25 R1: Longest Consecutive Sequence

Secondary task: Create edge-case tests: empty input, one element, duplicates, negatives, overflow.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Search in Rotated Sorted Array* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 27 (Nov 09), R3 Day 29 (Nov 11), R7 Day 33 (Nov 15), R14 Day 40 (Nov 22), R30 Day 56 (Dec 08), R60 Day 86 (Jan 07)

Day 27 — Week 4 — Monday, November 09, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Basic recursion tree / subsets preview

Learning objective: Understand the key invariant or decision rule for basic recursion tree / subsets preview; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Maximum Depth of Binary Tree — Easy; pattern: DFS recursion. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study basic recursion tree / subsets preview; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Maximum Depth of Binary Tree cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 13 R14: Binary Search; Day 20 R7: Merge Two Sorted Lists; Day 24 R3: Binary Search; Day 26 R1: Search in Rotated Sorted Array

Secondary task: Explain the pattern aloud in 3 minutes as if to an interviewer.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Maximum Depth of Binary Tree* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 28 (Nov 10), R3 Day 30 (Nov 12), R7 Day 34 (Nov 16), R14 Day 41 (Nov 23), R30 Day 57 (Dec 09), R60 Day 87 (Jan 08)

Day 28 — Week 4 — Tuesday, November 10, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Timed mixed arrays and strings

Learning objective: Understand the key invariant or decision rule for timed mixed arrays and strings; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Longest Consecutive Sequence — Medium; pattern: HashSet sequence starts. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering linked lists, stacks/queues, month 1 checkpoint; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Fast/slow pointers, intervals, monotonic structures.

Spaced-repetition queue due today: Day 14 R14: Merge Intervals; Day 21 R7: Linked List Cycle; Day 25 R3: Longest Consecutive Sequence; Day 27 R1: Maximum Depth of Binary Tree

Secondary task: Review Java pitfalls: equals/hashCode, nulls, String immutability, integer overflow.

Today's deliverable: Checkpoint: comprehensive cumulative assessment; compare with prior scores and write a 3-item remediation plan.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 29 (Nov 11), R3 Day 31 (Nov 13), R7 Day 35 (Nov 17), R14 Day 42 (Nov 24), R30 Day 58 (Dec 10), R60 Day 88 (Jan 09)

Week 4 checkpoint: Linked lists, stacks/queues, Month 1 checkpoint. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 29 — Week 5 — Wednesday, November 11, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Re-solve top misses from Month 1

Learning objective: Understand the key invariant or decision rule for re-solve top misses from month 1; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study re-solve top misses from month 1; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Counting Bits cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 15 R14: Product of Array Except Self; Day 22 R7: Unique Paths; Day 26 R3: Search in Rotated Sorted Array; Day 28 R1: Longest Consecutive Sequence

Secondary task: Add one mistake and one corrected test case to the error log.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Counting Bits* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 30 (Nov 12), R3 Day 32 (Nov 14), R7 Day 36 (Nov 18), R14 Day 43 (Nov 25), R30 Day 59 (Dec 11), R60 Day 89 (Jan 10)

Day 30 — Week 5 — Thursday, November 12, 2026

Phase: Java Foundations & Essential DSA **Total:** 3 hours (180 minutes)

Today's topic: Month 1 assessment and repair plan

Learning objective: Understand the key invariant or decision rule for month 1 assessment and repair plan; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study month 1 assessment and repair plan; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Median of Two Sorted Arrays cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 16 R14: Reverse Linked List; Day 23 R7: Two Sum; Day 27 R3: Maximum Depth of Binary Tree; Day 29 R1: Counting Bits

Secondary task: Implement a tiny Java example using ArrayList, HashMap, HashSet, and generics.

Today's deliverable: Monthly milestone: record independent solve rate, median time, CS/design confidence, project/recruiting evidence, and weakest three skills.

Completion criterion: Attempt *Median of Two Sorted Arrays* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 31 (Nov 13), R3 Day 33 (Nov 15), R7 Day 37 (Nov 19), R14 Day 44 (Nov 26), R30 Day 60 (Dec 12), R60 Day 90 (Jan 11)

Monthly assessment rubric: Score DSA independence (0–5), time/complexity (0–5), Java correctness (0–5), CS/design explanation (0–5), project evidence (0–5), behavioral clarity (0–5), recruiting consistency (0–5). Pass target: at least 4/5 in the month's core area and no critical gap; if below target, allocate three extra repair blocks in the next week and keep new content lighter.

Day 31 — Week 5 — Friday, November 13, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Fast/slow pointers: cycle detection

Learning objective: Understand the key invariant or decision rule for fast/slow pointers: cycle detection; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Merge Two Sorted Lists — Easy; pattern: dummy node. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study fast/slow pointers: cycle detection; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Merge Two Sorted Lists cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 1 R30: Longest Consecutive Sequence; Day 17 R14: Unique Paths; Day 24 R7: Binary Search; Day 28 R3: Longest Consecutive Sequence; Day 30 R1: Median of Two Sorted Arrays

Secondary task: Implement a reusable Java helper with clear invariants and null-safe behavior.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Merge Two Sorted Lists* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 32 (Nov 14), R3 Day 34 (Nov 16), R7 Day 38 (Nov 20), R14 Day 45 (Nov 27), R30 Day 61 (Dec 13), R60 Day 91 (Jan 12)

Day 32 — Week 5 — Saturday, November 14, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Intervals: merge and insert

Learning objective: Understand the key invariant or decision rule for intervals: merge and insert; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Binary Search — Easy; pattern: search invariant. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study intervals: merge and insert; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Binary Search cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 2 R30: Top K Frequent Elements; Day 18 R14: Linked List Cycle; Day 25 R7: Longest Consecutive Sequence; Day 29 R3: Counting Bits; Day 31 R1: Merge Two Sorted Lists

Secondary task: Compare two possible approaches and state why one is better for constraints.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Binary Search* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 33 (Nov 15), R3 Day 35 (Nov 17), R7 Day 39 (Nov 21), R14 Day 46 (Nov 28), R30 Day 62 (Dec 14), R60 Day 92 (Jan 13)

Day 33 — Week 5 — Sunday, November 15, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Monotonic stack: next greater element

Learning objective: Understand the key invariant or decision rule for monotonic stack: next greater element; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Valid Parentheses — Easy; pattern: stack. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study monotonic stack: next greater element; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Valid Parentheses cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 3 R30: Encode and Decode Strings; Day 19 R14: Daily Temperatures; Day 26 R7: Search in Rotated Sorted Array; Day 30 R3: Median of Two Sorted Arrays; Day 32 R1: Binary Search

Secondary task: Write tests for skewed trees, empty lists, duplicates, and boundary indices.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Valid Parentheses* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 34 (Nov 16), R3 Day 36 (Nov 18), R7 Day 40 (Nov 22), R14 Day 47 (Nov 29), R30 Day 63 (Dec 15), R60 Day 93 (Jan 14)

Day 34 — Week 5 — Monday, November 16, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Monotonic queue: sliding-window maximum

Learning objective: Understand the key invariant or decision rule for monotonic queue: sliding-window maximum; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Linked List Cycle — Easy; pattern: fast/slow pointers. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study monotonic queue: sliding-window maximum; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Linked List Cycle cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 4 R30: Product of Array Except Self; Day 20 R14: Merge Two Sorted Lists; Day 27 R7: Maximum Depth of Binary Tree; Day 31 R3: Merge Two Sorted Lists; Day 33 R1: Valid Parentheses

Secondary task: Practice explaining recursion stack space separately from auxiliary space.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Linked List Cycle* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 35 (Nov 17), R3 Day 37 (Nov 19), R7 Day 41 (Nov 23), R14 Day 48 (Nov 30), R30 Day 64 (Dec 16), R60 Day 94 (Jan 15)

Day 35 — Week 5 — Tuesday, November 17, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Binary tree recursion and invariants

Learning objective: Understand the key invariant or decision rule for binary tree recursion and invariants; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Binary Tree Level Order Traversal — Medium; pattern: BFS queue. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering fast/slow pointers, intervals, monotonic structures; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Trees, backtracking, heaps and tries.

Spaced-repetition queue due today: Day 5 R30: Single Number; Day 21 R14: Linked List Cycle; Day 28 R7: Longest Consecutive Sequence; Day 32 R3: Binary Search; Day 34 R1: Linked List Cycle

Secondary task: Add a short pattern card: trigger, invariant, complexity, common bug.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 36 (Nov 18), R3 Day 38 (Nov 20), R7 Day 42 (Nov 24), R14 Day 49 (Dec 01), R30 Day 65 (Dec 17), R60 Day 95 (Jan 16)

Week 5 checkpoint: Fast/slow pointers, intervals, monotonic structures. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 36 — Week 6 — Wednesday, November 18, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Tree DFS traversals

Learning objective: Understand the key invariant or decision rule for tree dfs traversals; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Binary Tree Maximum Path Sum — Hard; pattern: tree DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study tree dfs traversals; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Binary Tree Maximum Path Sum cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 6 R30: Valid Parentheses; Day 22 R14: Unique Paths; Day 29 R7: Counting Bits; Day 33 R3: Valid Parentheses; Day 35 R1: Binary Tree Level Order Traversal

Secondary task: Draw the recursion/decision tree and identify pruning opportunities.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Binary Tree Maximum Path Sum* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 37 (Nov 19), R3 Day 39 (Nov 21), R7 Day 43 (Nov 25), R14 Day 50 (Dec 02), R30 Day 66 (Dec 18), R60 Day 96 (Jan 17)

Day 37 — Week 6 — Thursday, November 19, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: pointer + stack patterns

Learning objective: Understand the key invariant or decision rule for weekly assessment: pointer + stack patterns; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Linked List Cycle — Easy; pattern: fast/slow pointers. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study weekly assessment: pointer + stack patterns; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Linked List Cycle cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 7 R30: Product of Array Except Self; Day 23 R14: Two Sum; Day 30 R7: Median of Two Sorted Arrays; Day 34 R3: Linked List Cycle; Day 36 R1: Binary Tree Maximum Path Sum

Secondary task: Implement a reusable Java helper with clear invariants and null-safe behavior.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Linked List Cycle* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 38 (Nov 20), R3 Day 40 (Nov 22), R7 Day 44 (Nov 26), R14 Day 51 (Dec 03), R30 Day 67 (Dec 19), R60 Day 97 (Jan 18)

Day 38 — Week 6 — Friday, November 20, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Tree BFS and level order

Learning objective: Understand the key invariant or decision rule for tree bfs and level order; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Binary Tree Maximum Path Sum — Hard; pattern: tree DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study tree bfs and level order; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Binary Tree Maximum Path Sum cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 8 R30: Group Anagrams; Day 24 R14: Binary Search; Day 31 R7: Merge Two Sorted Lists; Day 35 R3: Binary Tree Level Order Traversal; Day 37 R1: Linked List Cycle

Secondary task: Compare two possible approaches and state why one is better for constraints.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Binary Tree Maximum Path Sum* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 39 (Nov 21), R3 Day 41 (Nov 23), R7 Day 45 (Nov 27), R14 Day 52 (Dec 04), R30 Day 68 (Dec 20), R60 Day 98 (Jan 19)

Day 39 — Week 6 — Saturday, November 21, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: BST bounds and operations

Learning objective: Understand the key invariant or decision rule for bst bounds and operations; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Binary Tree Level Order Traversal — Medium; pattern: BFS queue. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study bst bounds and operations; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Binary Tree Level Order Traversal cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 9 R30: Longest Consecutive Sequence; Day 25 R14: Longest Consecutive Sequence; Day 32 R7: Binary Search; Day 36 R3: Binary Tree Maximum Path Sum; Day 38 R1: Binary Tree Maximum Path Sum

Secondary task: Write tests for skewed trees, empty lists, duplicates, and boundary indices.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Binary Tree Level Order Traversal* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 40 (Nov 22), R3 Day 42 (Nov 24), R7 Day 46 (Nov 28), R14 Day 53 (Dec 05), R30 Day 69 (Dec 21), R60 Day 99 (Jan 20)

Day 40 — Week 6 — Sunday, November 22, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Lowest common ancestor

Learning objective: Understand the key invariant or decision rule for lowest common ancestor; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study lowest common ancestor; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Single Number cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 10 R30: 3Sum; Day 26 R14: Search in Rotated Sorted Array; Day 33 R7: Valid Parentheses; Day 37 R3: Linked List Cycle; Day 39 R1: Binary Tree Level Order Traversal

Secondary task: Practice explaining recursion stack space separately from auxiliary space.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Single Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 41 (Nov 23), R3 Day 43 (Nov 25), R7 Day 47 (Nov 29), R14 Day 54 (Dec 06), R30 Day 70 (Dec 22), R60 Day 100 (Jan 21)

Day 41 — Week 6 — Monday, November 23, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Tree path sums and tree DP

Learning objective: Understand the key invariant or decision rule for tree path sums and tree dp; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Climbing Stairs — Easy; pattern: 1D DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study tree path sums and tree dp; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Climbing Stairs cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 11 R30: Minimum Window Substring; Day 27 R14: Maximum Depth of Binary Tree; Day 34 R7: Linked List Cycle; Day 38 R3: Binary Tree Maximum Path Sum; Day 40 R1: Single Number

Secondary task: Add a short pattern card: trigger, invariant, complexity, common bug.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Climbing Stairs* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 42 (Nov 24), R3 Day 44 (Nov 26), R7 Day 48 (Nov 30), R14 Day 55 (Dec 07), R30 Day 71 (Dec 23), R60 Day 101 (Jan 22)

Day 42 — Week 6 — Tuesday, November 24, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Backtracking decision trees

Learning objective: Understand the key invariant or decision rule for backtracking decision trees; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Maximum Depth of Binary Tree — Easy; pattern: DFS recursion. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering trees, backtracking, heaps and tries; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Greedy, matrices, timed core-pattern practice.

Spaced-repetition queue due today: Day 12 R30: Find Minimum in Rotated Sorted Array; Day 28 R14: Longest Consecutive Sequence; Day 35 R7: Binary Tree Level Order Traversal; Day 39 R3: Binary Tree Level Order Traversal; Day 41 R1: Climbing Stairs

Secondary task: Draw the recursion/decision tree and identify pruning opportunities.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 43 (Nov 25), R3 Day 45 (Nov 27), R7 Day 49 (Dec 01), R14 Day 56 (Dec 08), R30 Day 72 (Dec 24), R60 Day 102 (Jan 23)

Week 6 checkpoint: Trees, backtracking, heaps and tries. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 43 — Week 7 — Wednesday, November 25, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Subsets and combinations

Learning objective: Understand the key invariant or decision rule for subsets and combinations; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Kth Largest Element in an Array — Medium; pattern: heap / quickselect. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study subsets and combinations; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Kth Largest Element in an Array cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 13 R30: Binary Search; Day 29 R14: Counting Bits; Day 36 R7: Binary Tree Maximum Path Sum; Day 40 R3: Single Number; Day 42 R1: Maximum Depth of Binary Tree

Secondary task: Implement a reusable Java helper with clear invariants and null-safe behavior.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Kth Largest Element in an Array* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 44 (Nov 26), R3 Day 46 (Nov 28), R7 Day 50 (Dec 02), R14 Day 57 (Dec 09), R30 Day 73 (Dec 25), R60 Day 103 (Jan 24)

Day 44 — Week 7 — Thursday, November 26, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Holiday-aware adjustment: Thanksgiving: lighter active-recall and re-solve emphasis; keep all 180 minutes.

Today's topic: Weekly assessment: trees + backtracking

Learning objective: Understand the key invariant or decision rule for weekly assessment: trees + backtracking; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Maximum Depth of Binary Tree — Easy; pattern: DFS recursion. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study weekly assessment: trees + backtracking; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Maximum Depth of Binary Tree cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 14 R30: Merge Intervals; Day 30 R14: Median of Two Sorted Arrays; Day 37 R7: Linked List Cycle; Day 41 R3: Climbing Stairs; Day 43 R1: Kth Largest Element in an Array

Secondary task: Compare two possible approaches and state why one is better for constraints.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Maximum Depth of Binary Tree* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 45 (Nov 27), R3 Day 47 (Nov 29), R7 Day 51 (Dec 03), R14 Day 58 (Dec 10), R30 Day 74 (Dec 26), R60 Day 104 (Jan 25)

Day 45 — Week 7 — Friday, November 27, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Combination Sum and pruning

Learning objective: Understand the key invariant or decision rule for combination sum and pruning; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Kth Largest Element in an Array — Medium; pattern: heap / quickselect. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study combination sum and pruning; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Kth Largest Element in an Array cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 15 R30: Product of Array Except Self; Day 31 R14: Merge Two Sorted Lists; Day 38 R7: Binary Tree Maximum Path Sum; Day 42 R3: Maximum Depth of Binary Tree; Day 44 R1: Maximum Depth of Binary Tree

Secondary task: Write tests for skewed trees, empty lists, duplicates, and boundary indices.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Kth Largest Element in an Array* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 46 (Nov 28), R3 Day 48 (Nov 30), R7 Day 52 (Dec 04), R14 Day 59 (Dec 11), R30 Day 75 (Dec 27), R60 Day 105 (Jan 26)

Day 46 — Week 7 — Saturday, November 28, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Grid DFS / Word Search

Learning objective: Understand the key invariant or decision rule for grid dfs / word search; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Subsets — Medium; pattern: backtracking decision tree. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study grid dfs / word search; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Subsets cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 16 R30: Reverse Linked List; Day 32 R14: Binary Search; Day 39 R7: Binary Tree Level Order Traversal; Day 43 R3: Kth Largest Element in an Array; Day 45 R1: Kth Largest Element in an Array

Secondary task: Practice explaining recursion stack space separately from auxiliary space.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Subsets* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 47 (Nov 29), R3 Day 49 (Dec 01), R7 Day 53 (Dec 05), R14 Day 60 (Dec 12), R30 Day 76 (Dec 28), R60 Day 106 (Jan 27)

Day 47 — Week 7 — Sunday, November 29, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Heap and PriorityQueue

Learning objective: Understand the key invariant or decision rule for heap and priorityqueue; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Kth Largest Element in an Array — Medium; pattern: heap / quickselect. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study heap and priorityqueue; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Kth Largest Element in an Array cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 17 R30: Unique Paths; Day 33 R14: Valid Parentheses; Day 40 R7: Single Number; Day 44 R3: Maximum Depth of Binary Tree; Day 46 R1: Subsets

Secondary task: Add a short pattern card: trigger, invariant, complexity, common bug.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Kth Largest Element in an Array* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 48 (Nov 30), R3 Day 50 (Dec 02), R7 Day 54 (Dec 06), R14 Day 61 (Dec 13), R30 Day 77 (Dec 29), R60 Day 107 (Jan 28)

Day 48 — Week 7 — Monday, November 30, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Top-K patterns

Learning objective: Understand the key invariant or decision rule for top-k patterns; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Subsets — Medium; pattern: backtracking decision tree. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study top-k patterns; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Subsets cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 18 R30: Linked List Cycle; Day 34 R14: Linked List Cycle; Day 41 R7: Climbing Stairs; Day 45 R3: Kth Largest Element in an Array; Day 47 R1: Kth Largest Element in an Array

Secondary task: Draw the recursion/decision tree and identify pruning opportunities.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Subsets* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 49 (Dec 01), R3 Day 51 (Dec 03), R7 Day 55 (Dec 07), R14 Day 62 (Dec 14), R30 Day 78 (Dec 30), R60 Day 108 (Jan 29)

Day 49 — Week 7 — Tuesday, December 01, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Trie implementation

Learning objective: Understand the key invariant or decision rule for trie implementation; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Task Scheduler — Medium; pattern: greedy counting. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering greedy, matrices, timed core-pattern practice; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Mixed core patterns and Month 2 checkpoint.

Spaced-repetition queue due today: Day 19 R30: Daily Temperatures; Day 35 R14: Binary Tree Level Order Traversal; Day 42 R7: Maximum Depth of Binary Tree; Day 46 R3: Subsets; Day 48 R1: Subsets

Secondary task: Implement a reusable Java helper with clear invariants and null-safe behavior.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 50 (Dec 02), R3 Day 52 (Dec 04), R7 Day 56 (Dec 08), R14 Day 63 (Dec 15), R30 Day 79 (Dec 31), R60 Day 109 (Jan 30)

Week 7 checkpoint: Greedy, matrices, timed core-pattern practice. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 50 — Week 8 — Wednesday, December 02, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Greedy scheduling and proof intuition

Learning objective: Understand the key invariant or decision rule for greedy scheduling and proof intuition; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Combination Sum — Medium; pattern: backtracking + pruning. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study greedy scheduling and proof intuition; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Combination Sum cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 20 R30: Merge Two Sorted Lists; Day 36 R14: Binary Tree Maximum Path Sum; Day 43 R7: Kth Largest Element in an Array; Day 47 R3: Kth Largest Element in an Array; Day 49 R1: Task Scheduler

Secondary task: Compare two possible approaches and state why one is better for constraints.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Combination Sum* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 51 (Dec 03), R3 Day 53 (Dec 05), R7 Day 57 (Dec 09), R14 Day 64 (Dec 16), R30 Day 80 (Jan 01), R60 Day 110 (Jan 31)

Day 51 — Week 8 — Thursday, December 03, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: heaps + greedy

Learning objective: Understand the key invariant or decision rule for weekly assessment: heaps + greedy; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Task Scheduler — Medium; pattern: greedy counting. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study weekly assessment: heaps + greedy; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Task Scheduler cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 21 R30: Linked List Cycle; Day 37 R14: Linked List Cycle; Day 44 R7: Maximum Depth of Binary Tree; Day 48 R3: Subsets; Day 50 R1: Combination Sum

Secondary task: Write tests for skewed trees, empty lists, duplicates, and boundary indices.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Task Scheduler* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 52 (Dec 04), R3 Day 54 (Dec 06), R7 Day 58 (Dec 10), R14 Day 65 (Dec 17), R30 Day 81 (Jan 02), R60 Day 111 (Feb 01)

Day 52 — Week 8 — Friday, December 04, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Matrix traversal and flood fill

Learning objective: Understand the key invariant or decision rule for matrix traversal and flood fill; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Koko Eating Bananas — Medium; pattern: binary search on answer. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study matrix traversal and flood fill; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Koko Eating Bananas cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 22 R30: Unique Paths; Day 38 R14: Binary Tree Maximum Path Sum; Day 45 R7: Kth Largest Element in an Array; Day 49 R3: Task Scheduler; Day 51 R1: Task Scheduler

Secondary task: Practice explaining recursion stack space separately from auxiliary space.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Koko Eating Bananas* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 53 (Dec 05), R3 Day 55 (Dec 07), R7 Day 59 (Dec 11), R14 Day 66 (Dec 18), R30 Day 82 (Jan 03), R60 Day 112 (Feb 02)

Day 53 — Week 8 — Saturday, December 05, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Rotated binary search

Learning objective: Understand the key invariant or decision rule for rotated binary search; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search in Rotated Sorted Array — Medium; pattern: modified binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study rotated binary search; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Search in Rotated Sorted Array cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 23 R30: Two Sum; Day 39 R14: Binary Tree Level Order Traversal; Day 46 R7: Subsets; Day 50 R3: Combination Sum; Day 52 R1: Koko Eating Bananas

Secondary task: Add a short pattern card: trigger, invariant, complexity, common bug.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Search in Rotated Sorted Array* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 54 (Dec 06), R3 Day 56 (Dec 08), R7 Day 60 (Dec 12), R14 Day 67 (Dec 19), R30 Day 83 (Jan 04), R60 Day 113 (Feb 03)

Day 54 — Week 8 — Sunday, December 06, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Binary search on answer

Learning objective: Understand the key invariant or decision rule for binary search on answer; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Merge Intervals — Medium; pattern: sort + sweep. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study binary search on answer; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Merge Intervals cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 24 R30: Binary Search; Day 40 R14: Single Number; Day 47 R7: Kth Largest Element in an Array; Day 51 R3: Task Scheduler; Day 53 R1: Search in Rotated Sorted Array

Secondary task: Draw the recursion/decision tree and identify pruning opportunities.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Merge Intervals* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 55 (Dec 07), R3 Day 57 (Dec 09), R7 Day 61 (Dec 13), R14 Day 68 (Dec 20), R30 Day 84 (Jan 05), R60 Day 114 (Feb 04)

Day 55 — Week 8 — Monday, December 07, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Intervals mixed set

Learning objective: Understand the key invariant or decision rule for intervals mixed set; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find Minimum in Rotated Sorted Array — Medium; pattern: partition invariant. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study intervals mixed set; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Find Minimum in Rotated Sorted Array cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 25 R30: Longest Consecutive Sequence; Day 41 R14: Climbing Stairs; Day 48 R7: Subsets; Day 52 R3: Koko Eating Bananas; Day 54 R1: Merge Intervals

Secondary task: Implement a reusable Java helper with clear invariants and null-safe behavior.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Find Minimum in Rotated Sorted Array* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 56 (Dec 08), R3 Day 58 (Dec 10), R7 Day 62 (Dec 14), R14 Day 69 (Dec 21), R30 Day 85 (Jan 06), R60 Day 115 (Feb 05)

Day 56 — Week 8 — Tuesday, December 08, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Backtracking mixed timed practice

Learning objective: Understand the key invariant or decision rule for backtracking mixed timed practice; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Word Search — Medium; pattern: grid DFS + backtracking. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering mixed core patterns and month 2 checkpoint; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Graph representations, DFS/BFS, cycle detection.

Spaced-repetition queue due today: Day 26 R30: Search in Rotated Sorted Array; Day 42 R14: Maximum Depth of Binary Tree; Day 49 R7: Task Scheduler; Day 53 R3: Search in Rotated Sorted Array; Day 55 R1: Find Minimum in Rotated Sorted Array

Secondary task: Compare two possible approaches and state why one is better for constraints.

Today's deliverable: Checkpoint: comprehensive cumulative assessment; compare with prior scores and write a 3-item remediation plan.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 57 (Dec 09), R3 Day 59 (Dec 11), R7 Day 63 (Dec 15), R14 Day 70 (Dec 22), R30 Day 86 (Jan 07), R60 Day 116 (Feb 06)

Week 8 checkpoint: Mixed core patterns and Month 2 checkpoint. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 57 — Week 9 — Wednesday, December 09, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Trees mixed timed practice

Learning objective: Understand the key invariant or decision rule for trees mixed timed practice; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Binary Tree Maximum Path Sum — Hard; pattern: tree DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study trees mixed timed practice; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Binary Tree Maximum Path Sum cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 27 R30: Maximum Depth of Binary Tree; Day 43 R14: Kth Largest Element in an Array; Day 50 R7: Combination Sum; Day 54 R3: Merge Intervals; Day 56 R1: Word Search

Secondary task: Write tests for skewed trees, empty lists, duplicates, and boundary indices.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Binary Tree Maximum Path Sum* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 58 (Dec 10), R3 Day 60 (Dec 12), R7 Day 64 (Dec 16), R14 Day 71 (Dec 23), R30 Day 87 (Jan 08), R60 Day 117 (Feb 07)

Day 58 — Week 9 — Thursday, December 10, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Mock coding round 1

Learning objective: Understand the key invariant or decision rule for mock coding round 1; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study mock coding round 1; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Find the Duplicate Number cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 28 R30: Longest Consecutive Sequence; Day 44 R14: Maximum Depth of Binary Tree; Day 51 R7: Task Scheduler; Day 55 R3: Find Minimum in Rotated Sorted Array; Day 57 R1: Binary Tree Maximum Path Sum

Secondary task: Practice explaining recursion stack space separately from auxiliary space.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Find the Duplicate Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 59 (Dec 11), R3 Day 61 (Dec 13), R7 Day 65 (Dec 17), R14 Day 72 (Dec 24), R30 Day 88 (Jan 09), R60 Day 118 (Feb 08)

Day 59 — Week 9 — Friday, December 11, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Repair weak patterns

Learning objective: Understand the key invariant or decision rule for repair weak patterns; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study repair weak patterns; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Single Number cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 29 R30: Counting Bits; Day 45 R14: Kth Largest Element in an Array; Day 52 R7: Koko Eating Bananas; Day 56 R3: Word Search; Day 58 R1: Find the Duplicate Number

Secondary task: Add a short pattern card: trigger, invariant, complexity, common bug.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Single Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 60 (Dec 12), R3 Day 62 (Dec 14), R7 Day 66 (Dec 18), R14 Day 73 (Dec 25), R30 Day 89 (Jan 10), R60 Day 119 (Feb 09)

Day 60 — Week 9 — Saturday, December 12, 2026

Phase: Core Interview Patterns **Total:** 3 hours (180 minutes)

Today's topic: Month 2 assessment and review

Learning objective: Understand the key invariant or decision rule for month 2 assessment and review; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study month 2 assessment and review; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Search a 2D Matrix cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 30 R30: Median of Two Sorted Arrays; Day 46 R14: Subsets; Day 53 R7: Search in Rotated Sorted Array; Day 57 R3: Binary Tree Maximum Path Sum; Day 59 R1: Single Number

Secondary task: Draw the recursion/decision tree and identify pruning opportunities.

Today's deliverable: Monthly milestone: record independent solve rate, median time, CS/design confidence, project/recruiting evidence, and weakest three skills.

Completion criterion: Attempt *Search a 2D Matrix* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 61 (Dec 13), R3 Day 63 (Dec 15), R7 Day 67 (Dec 19), R14 Day 74 (Dec 26), R30 Day 90 (Jan 11), R60 Day 120 (Feb 10)

Monthly assessment rubric: Score DSA independence (0–5), time/complexity (0–5), Java correctness (0–5), CS/design explanation (0–5), project evidence (0–5), behavioral clarity (0–5), recruiting consistency (0–5). Pass target: at least 4/5 in the month's core area and no critical gap; if below target, allocate three extra repair blocks in the next week and keep new content lighter.

Day 61 — Week 9 — Sunday, December 13, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Graph representations: adjacency list/matrix

Learning objective: Understand the key invariant or decision rule for graph representations: adjacency list/matrix; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Course Schedule — Medium; pattern: cycle detection / topological sort. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study graph representations: adjacency list/matrix; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Course Schedule cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 1 R60: Longest Consecutive Sequence; Day 31 R30: Merge Two Sorted Lists; Day 47 R14: Kth Largest Element in an Array; Day 54 R7: Merge Intervals; Day 58 R3: Find the Duplicate Number; Day 60 R1: Search a 2D Matrix

Secondary task: Explain why the algorithm is correct and when it fails (e.g., negative edges for Dijkstra).

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Course Schedule* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 62 (Dec 14), R3 Day 64 (Dec 16), R7 Day 68 (Dec 20), R14 Day 75 (Dec 27), R30 Day 91 (Jan 12), R60 Day 121 (Feb 11)

Day 62 — Week 9 — Monday, December 14, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Graph DFS and visited-state design

Learning objective: Understand the key invariant or decision rule for graph dfs and visited-state design; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Network Delay Time — Medium; pattern: Dijkstra. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study graph dfs and visited-state design; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Network Delay Time cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 2 R60: Top K Frequent Elements; Day 32 R30: Binary Search; Day 48 R14: Subsets; Day 55 R7: Find Minimum in Rotated Sorted Array; Day 59 R3: Single Number; Day 61 R1: Course Schedule

Secondary task: Implement with adjacency lists, ArrayDeque, PriorityQueue, or arrays as appropriate.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Network Delay Time* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 63 (Dec 15), R3 Day 65 (Dec 17), R7 Day 69 (Dec 21), R14 Day 76 (Dec 28), R30 Day 92 (Jan 13), R60 Day 122 (Feb 12)

Day 63 — Week 9 — Tuesday, December 15, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Graph BFS and shortest unweighted paths

Learning objective: Understand the key invariant or decision rule for graph bfs and shortest unweighted paths; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Course Schedule — Medium; pattern: cycle detection / topological sort. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering graph representations, dfs/bfs, cycle detection; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Topological sort, Union-Find, shortest paths.

Spaced-repetition queue due today: Day 3 R60: Encode and Decode Strings; Day 33 R30: Valid Parentheses; Day 49 R14: Task Scheduler; Day 56 R7: Word Search; Day 60 R3: Search a 2D Matrix; Day 62 R1: Network Delay Time

Secondary task: Test disconnected graphs, cycles, duplicate edges, empty input, and unreachable nodes.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 64 (Dec 16), R3 Day 66 (Dec 18), R7 Day 70 (Dec 22), R14 Day 77 (Dec 29), R30 Day 93 (Jan 14), R60 Day 123 (Feb 13)

Week 9 checkpoint: Graph representations, DFS/BFS, cycle detection. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 64 — Week 10 — Wednesday, December 16, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Grid graph: Number of Islands

Learning objective: Understand the key invariant or decision rule for grid graph: number of islands; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Network Delay Time — Medium; pattern: Dijkstra. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study grid graph: number of islands; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Network Delay Time cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 4 R60: Product of Array Except Self; Day 34 R30: Linked List Cycle; Day 50 R14: Combination Sum; Day 57 R7: Binary Tree Maximum Path Sum; Day 61 R3: Course Schedule; Day 63 R1: Course Schedule

Secondary task: Compare brute force to optimized complexity and justify the trade-off.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Network Delay Time* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 65 (Dec 17), R3 Day 67 (Dec 19), R7 Day 71 (Dec 23), R14 Day 78 (Dec 30), R30 Day 94 (Jan 15), R60 Day 124 (Feb 14)

Day 65 — Week 10 — Thursday, December 17, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Cycle detection in directed graphs

Learning objective: Understand the key invariant or decision rule for cycle detection in directed graphs; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Clone Graph — Medium; pattern: DFS/BFS + map. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study cycle detection in directed graphs; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Clone Graph cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 5 R60: Single Number; Day 35 R30: Binary Tree Level Order Traversal; Day 51 R14: Task Scheduler; Day 58 R7: Find the Duplicate Number; Day 62 R3: Network Delay Time; Day 64 R1: Network Delay Time

Secondary task: Add a pattern card and one counterexample to your notes.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Clone Graph* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 66 (Dec 18), R3 Day 68 (Dec 20), R7 Day 72 (Dec 24), R14 Day 79 (Dec 31), R30 Day 95 (Jan 16), R60 Day 125 (Feb 15)

Day 66 — Week 10 — Friday, December 18, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Topological sorting / prerequisites

Learning objective: Understand the key invariant or decision rule for topological sorting / prerequisites; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Redundant Connection — Medium; pattern: Union-Find. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study topological sorting / prerequisites; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Redundant Connection cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 6 R60: Valid Parentheses; Day 36 R30: Binary Tree Maximum Path Sum; Day 52 R14: Koko Eating Bananas; Day 59 R7: Single Number; Day 63 R3: Course Schedule; Day 65 R1: Clone Graph

Secondary task: Draw the graph/DP state model before coding; label transitions clearly.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Redundant Connection* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 67 (Dec 19), R3 Day 69 (Dec 21), R7 Day 73 (Dec 25), R14 Day 80 (Jan 01), R30 Day 96 (Jan 17), R60 Day 126 (Feb 16)

Day 67 — Week 10 — Saturday, December 19, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: graph traversal

Learning objective: Understand the key invariant or decision rule for weekly assessment: graph traversal; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Number of Islands — Medium; pattern: grid DFS/BFS. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study weekly assessment: graph traversal; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Number of Islands cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 7 R60: Product of Array Except Self; Day 37 R30: Linked List Cycle; Day 53 R14: Search in Rotated Sorted Array; Day 60 R7: Search a 2D Matrix; Day 64 R3: Network Delay Time; Day 66 R1: Redundant Connection

Secondary task: Explain why the algorithm is correct and when it fails (e.g., negative edges for Dijkstra).

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Number of Islands* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 68 (Dec 20), R3 Day 70 (Dec 22), R7 Day 74 (Dec 26), R14 Day 81 (Jan 02), R30 Day 97 (Jan 18), R60 Day 127 (Feb 17)

Day 68 — Week 10 — Sunday, December 20, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Union-Find: parent/rank/path compression

Learning objective: Understand the key invariant or decision rule for union-find: parent/rank/path compression; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Rotting Oranges — Medium; pattern: multi-source BFS. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study union-find: parent/rank/path compression; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Rotting Oranges cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 8 R60: Group Anagrams; Day 38 R30: Binary Tree Maximum Path Sum; Day 54 R14: Merge Intervals; Day 61 R7: Course Schedule; Day 65 R3: Clone Graph; Day 67 R1: Number of Islands

Secondary task: Implement with adjacency lists, ArrayDeque, PriorityQueue, or arrays as appropriate.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Rotting Oranges* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 69 (Dec 21), R3 Day 71 (Dec 23), R7 Day 75 (Dec 27), R14 Day 82 (Jan 03), R30 Day 98 (Jan 19), R60 Day 128 (Feb 18)

Day 69 — Week 10 — Monday, December 21, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Dijkstra and nonnegative edge weights

Learning objective: Understand the key invariant or decision rule for dijkstra and nonnegative edge weights; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Min Cost to Connect All Points — Medium; pattern: minimum spanning tree. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study dijkstra and nonnegative edge weights; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Min Cost to Connect All Points cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 9 R60: Longest Consecutive Sequence; Day 39 R30: Binary Tree Level Order Traversal; Day 55 R14: Find Minimum in Rotated Sorted Array; Day 62 R7: Network Delay Time; Day 66 R3: Redundant Connection; Day 68 R1: Rotting Oranges

Secondary task: Test disconnected graphs, cycles, duplicate edges, empty input, and unreachable nodes.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Min Cost to Connect All Points* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 70 (Dec 22), R3 Day 72 (Dec 24), R7 Day 76 (Dec 28), R14 Day 83 (Jan 04), R30 Day 99 (Jan 20), R60 Day 129 (Feb 19)

Day 70 — Week 10 — Tuesday, December 22, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Minimum spanning tree intuition

Learning objective: Understand the key invariant or decision rule for minimum spanning tree intuition; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Rotting Oranges — Medium; pattern: multi-source BFS. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering topological sort, union-find, shortest paths; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: DP state design, 1D/2D DP, knapsack.

Spaced-repetition queue due today: Day 10 R60: 3Sum; Day 40 R30: Single Number; Day 56 R14: Word Search; Day 63 R7: Course Schedule; Day 67 R3: Number of Islands; Day 69 R1: Min Cost to Connect All Points

Secondary task: Compare brute force to optimized complexity and justify the trade-off.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 71 (Dec 23), R3 Day 73 (Dec 25), R7 Day 77 (Dec 29), R14 Day 84 (Jan 05), R30 Day 100 (Jan 21), R60 Day 130 (Feb 20)

Week 10 checkpoint: Topological sort, Union-Find, shortest paths. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 71 — Week 11 — Wednesday, December 23, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Bellman-Ford and negative edges overview

Learning objective: Understand the key invariant or decision rule for bellman-ford and negative edges overview; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study bellman-ford and negative edges overview; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Search a 2D Matrix cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 11 R60: Minimum Window Substring; Day 41 R30: Climbing Stairs; Day 57 R14: Binary Tree Maximum Path Sum; Day 64 R7: Network Delay Time; Day 68 R3: Rotting Oranges; Day 70 R1: Rotting Oranges

Secondary task: Add a pattern card and one counterexample to your notes.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Search a 2D Matrix* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 72 (Dec 24), R3 Day 74 (Dec 26), R7 Day 78 (Dec 30), R14 Day 85 (Jan 06), R30 Day 101 (Jan 22), R60 Day 131 (Feb 21)

Day 72 — Week 11 — Thursday, December 24, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Holiday-aware adjustment: Christmas Eve: lighter revision, notes cleanup and low-pressure practice; keep all 180 minutes.

Today's topic: 1D DP: state and recurrence

Learning objective: Understand the key invariant or decision rule for 1d dp: state and recurrence; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Coin Change — Medium; pattern: unbounded knapsack. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study 1d dp: state and recurrence; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Coin Change cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 12 R60: Find Minimum in Rotated Sorted Array; Day 42 R30: Maximum Depth of Binary Tree; Day 58 R14: Find the Duplicate Number; Day 65 R7: Clone Graph; Day 69 R3: Min Cost to Connect All Points; Day 71 R1: Search a 2D Matrix

Secondary task: Draw the graph/DP state model before coding; label transitions clearly.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Coin Change* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 73 (Dec 25), R3 Day 75 (Dec 27), R7 Day 79 (Dec 31), R14 Day 86 (Jan 07), R30 Day 102 (Jan 23), R60 Day 132 (Feb 22)

Day 73 — Week 11 — Friday, December 25, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Holiday-aware adjustment: Christmas Day: use a calm 3-hour review/project-notes session; do not skip the commitment.

Today's topic: Climbing Stairs / House Robber

Learning objective: Understand the key invariant or decision rule for climbing stairs / house robber; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Unique Paths — Medium; pattern: grid DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study climbing stairs / house robber; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Unique Paths cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 13 R60: Binary Search; Day 43 R30: Kth Largest Element in an Array; Day 59 R14: Single Number; Day 66 R7: Redundant Connection; Day 70 R3: Rotting Oranges; Day 72 R1: Coin Change

Secondary task: Explain why the algorithm is correct and when it fails (e.g., negative edges for Dijkstra).

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Unique Paths* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 74 (Dec 26), R3 Day 76 (Dec 28), R7 Day 80 (Jan 01), R14 Day 87 (Jan 08), R30 Day 103 (Jan 24), R60 Day 133 (Feb 23)

Day 74 — Week 11 — Saturday, December 26, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: graph + DP

Learning objective: Understand the key invariant or decision rule for weekly assessment: graph + dp; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Clone Graph — Medium; pattern: DFS/BFS + map. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study weekly assessment: graph + dp; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Clone Graph cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 14 R60: Merge Intervals; Day 44 R30: Maximum Depth of Binary Tree; Day 60 R14: Search a 2D Matrix; Day 67 R7: Number of Islands; Day 71 R3: Search a 2D Matrix; Day 73 R1: Unique Paths

Secondary task: Implement with adjacency lists, ArrayDeque, PriorityQueue, or arrays as appropriate.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Clone Graph* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 75 (Dec 27), R3 Day 77 (Dec 29), R7 Day 81 (Jan 02), R14 Day 88 (Jan 09), R30 Day 104 (Jan 25), R60 Day 134 (Feb 24)

Day 75 — Week 11 — Sunday, December 27, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Coin Change and knapsack patterns

Learning objective: Understand the key invariant or decision rule for coin change and knapsack patterns; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Longest Increasing Subsequence — Medium; pattern: DP / binary-search optimization. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study coin change and knapsack patterns; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Longest Increasing Subsequence cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 15 R60: Product of Array Except Self; Day 45 R30: Kth Largest Element in an Array; Day 61 R14: Course Schedule; Day 68 R7: Rotting Oranges; Day 72 R3: Coin Change; Day 74 R1: Clone Graph

Secondary task: Test disconnected graphs, cycles, duplicate edges, empty input, and unreachable nodes.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Longest Increasing Subsequence* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 76 (Dec 28), R3 Day 78 (Dec 30), R7 Day 82 (Jan 03), R14 Day 89 (Jan 10), R30 Day 105 (Jan 26), R60 Day 135 (Feb 25)

Day 76 — Week 11 — Monday, December 28, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Grid DP and Unique Paths

Learning objective: Understand the key invariant or decision rule for grid dp and unique paths; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Word Break — Medium; pattern: prefix DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study grid dp and unique paths; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Word Break cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 16 R60: Reverse Linked List; Day 46 R30: Subsets; Day 62 R14: Network Delay Time; Day 69 R7: Min Cost to Connect All Points; Day 73 R3: Unique Paths; Day 75 R1: Longest Increasing Subsequence

Secondary task: Compare brute force to optimized complexity and justify the trade-off.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Word Break* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 77 (Dec 29), R3 Day 79 (Dec 31), R7 Day 83 (Jan 04), R14 Day 90 (Jan 11), R30 Day 106 (Jan 27), R60 Day 136 (Feb 26)

Day 77 — Week 11 — Tuesday, December 29, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: LCS and subsequence DP

Learning objective: Understand the key invariant or decision rule for lcs and subsequence dp; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Coin Change — Medium; pattern: unbounded knapsack. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering dp state design, 1d/2d dp, knapsack; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Subsequences, bit manipulation, mixed advanced DSA.

Spaced-repetition queue due today: Day 17 R60: Unique Paths; Day 47 R30: Kth Largest Element in an Array; Day 63 R14: Course Schedule; Day 70 R7: Rotting Oranges; Day 74 R3: Clone Graph; Day 76 R1: Word Break

Secondary task: Add a pattern card and one counterexample to your notes.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 78 (Dec 30), R3 Day 80 (Jan 01), R7 Day 84 (Jan 05), R14 Day 91 (Jan 12), R30 Day 107 (Jan 28), R60 Day 137 (Feb 27)

Week 11 checkpoint: DP state design, 1D/2D DP, knapsack. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 78 — Week 12 — Wednesday, December 30, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: LIS and optimization trade-offs

Learning objective: Understand the key invariant or decision rule for lis and optimization trade-offs; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Unique Paths — Medium; pattern: grid DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study lis and optimization trade-offs; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Unique Paths cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 18 R60: Linked List Cycle; Day 48 R30: Subsets; Day 64 R14: Network Delay Time; Day 71 R7: Search a 2D Matrix; Day 75 R3: Longest Increasing Subsequence; Day 77 R1: Coin Change

Secondary task: Draw the graph/DP state model before coding; label transitions clearly.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Unique Paths* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 79 (Dec 31), R3 Day 81 (Jan 02), R7 Day 85 (Jan 06), R14 Day 92 (Jan 13), R30 Day 108 (Jan 29), R60 Day 138 (Feb 28)

Day 79 — Week 12 — Thursday, December 31, 2026

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Holiday-aware adjustment: New Year's Eve: lighter cumulative recall and planning; keep all 180 minutes.

Today's topic: Word Break / prefix DP

Learning objective: Understand the key invariant or decision rule for word break / prefix dp; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Climbing Stairs — Easy; pattern: 1D DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study word break / prefix dp; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Climbing Stairs cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 19 R60: Daily Temperatures; Day 49 R30: Task Scheduler; Day 65 R14: Clone Graph; Day 72 R7: Coin Change; Day 76 R3: Word Break; Day 78 R1: Unique Paths

Secondary task: Explain why the algorithm is correct and when it fails (e.g., negative edges for Dijkstra).

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Climbing Stairs* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 80 (Jan 01), R3 Day 82 (Jan 03), R7 Day 86 (Jan 07), R14 Day 93 (Jan 14), R30 Day 109 (Jan 30), R60 Day 139 (Mar 01)

Day 80 — Week 12 — Friday, January 01, 2027

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Holiday-aware adjustment: New Year's Day: use a lighter review session and set weekly goals; keep all 180 minutes.

Today's topic: Edit Distance / 2D transitions

Learning objective: Understand the key invariant or decision rule for edit distance / 2d transitions; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Longest Increasing Subsequence — Medium; pattern: DP / binary-search optimization. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study edit distance / 2d transitions; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Longest Increasing Subsequence cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 20 R60: Merge Two Sorted Lists; Day 50 R30: Combination Sum; Day 66 R14: Redundant Connection; Day 73 R7: Unique Paths; Day 77 R3: Coin Change; Day 79 R1: Climbing Stairs

Secondary task: Implement with adjacency lists, ArrayDeque, PriorityQueue, or arrays as appropriate.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Longest Increasing Subsequence* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 81 (Jan 02), R3 Day 83 (Jan 04), R7 Day 87 (Jan 08), R14 Day 94 (Jan 15), R30 Day 110 (Jan 31), R60 Day 140 (Mar 02)

Day 81 — Week 12 — Saturday, January 02, 2027

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: DP

Learning objective: Understand the key invariant or decision rule for weekly assessment: dp; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Word Break — Medium; pattern: prefix DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study weekly assessment: dp; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Word Break cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 21 R60: Linked List Cycle; Day 51 R30: Task Scheduler; Day 67 R14: Number of Islands; Day 74 R7: Clone Graph; Day 78 R3: Unique Paths; Day 80 R1: Longest Increasing Subsequence

Secondary task: Test disconnected graphs, cycles, duplicate edges, empty input, and unreachable nodes.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Word Break* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 82 (Jan 03), R3 Day 84 (Jan 05), R7 Day 88 (Jan 09), R14 Day 95 (Jan 16), R30 Day 111 (Feb 01), R60 Day 141 (Mar 03)

Day 82 — Week 12 — Sunday, January 03, 2027

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Bit manipulation: XOR, masks, shifts

Learning objective: Understand the key invariant or decision rule for bit manipulation: xor, masks, shifts; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study bit manipulation: xor, masks, shifts; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Find the Duplicate Number cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 22 R60: Unique Paths; Day 52 R30: Koko Eating Bananas; Day 68 R14: Rotting Oranges; Day 75 R7: Longest Increasing Subsequence; Day 79 R3: Climbing Stairs; Day 81 R1: Word Break

Secondary task: Compare brute force to optimized complexity and justify the trade-off.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Find the Duplicate Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 83 (Jan 04), R3 Day 85 (Jan 06), R7 Day 89 (Jan 10), R14 Day 96 (Jan 17), R30 Day 112 (Feb 02), R60 Day 142 (Mar 04)

Day 83 — Week 12 — Monday, January 04, 2027

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Advanced binary search partitions

Learning objective: Understand the key invariant or decision rule for advanced binary search partitions; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Binary Search — Easy; pattern: search invariant. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study advanced binary search partitions; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Binary Search cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 23 R60: Two Sum; Day 53 R30: Search in Rotated Sorted Array; Day 69 R14: Min Cost to Connect All Points; Day 76 R7: Word Break; Day 80 R3: Longest Increasing Subsequence; Day 82 R1: Find the Duplicate Number

Secondary task: Add a pattern card and one counterexample to your notes.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Binary Search* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 84 (Jan 05), R3 Day 86 (Jan 07), R7 Day 90 (Jan 11), R14 Day 97 (Jan 18), R30 Day 113 (Feb 03), R60 Day 143 (Mar 05)

Day 84 — Week 12 — Tuesday, January 05, 2027

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Mixed graph/DP timed set

Learning objective: Understand the key invariant or decision rule for mixed graph/dp timed set; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Network Delay Time — Medium; pattern: Dijkstra. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering subsequences, bit manipulation, mixed advanced dsa; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: OOP/LLD, OS, databases, networking.

Spaced-repetition queue due today: Day 24 R60: Binary Search; Day 54 R30: Merge Intervals; Day 70 R14: Rotting Oranges; Day 77 R7: Coin Change; Day 81 R3: Word Break; Day 83 R1: Binary Search

Secondary task: Draw the graph/DP state model before coding; label transitions clearly.

Today's deliverable: Checkpoint: comprehensive cumulative assessment; compare with prior scores and write a 3-item remediation plan.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 85 (Jan 06); R3 Day 87 (Jan 08); R7 Day 91 (Jan 12); R14 Day 98 (Jan 19); R30 Day 114 (Feb 04); R60 Day 144 (Mar 06)

Week 12 checkpoint: Subsequences, bit manipulation, mixed advanced DSA. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 85 — Week 13 — Wednesday, January 06, 2027

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Hard problem decomposition

Learning objective: Understand the key invariant or decision rule for hard problem decomposition; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study hard problem decomposition; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Find the Duplicate Number cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 25 R60: Longest Consecutive Sequence; Day 55 R30: Find Minimum in Rotated Sorted Array; Day 71 R14: Search a 2D Matrix; Day 78 R7: Unique Paths; Day 82 R3: Find the Duplicate Number; Day 84 R1: Network Delay Time

Secondary task: Explain why the algorithm is correct and when it fails (e.g., negative edges for Dijkstra).

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Find the Duplicate Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 86 (Jan 07), R3 Day 88 (Jan 09), R7 Day 92 (Jan 13), R14 Day 99 (Jan 20), R30 Day 115 (Feb 05), R60 Day 145 (Mar 07)

Day 86 — Week 13 — Thursday, January 07, 2027

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Brute force to optimized comparison

Learning objective: Understand the key invariant or decision rule for brute force to optimized comparison; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study brute force to optimized comparison; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Single Number cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 26 R60: Search in Rotated Sorted Array; Day 56 R30: Word Search; Day 72 R14: Coin Change; Day 79 R7: Climbing Stairs; Day 83 R3: Binary Search; Day 85 R1: Find the Duplicate Number

Secondary task: Implement with adjacency lists, ArrayDeque, PriorityQueue, or arrays as appropriate.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Single Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 87 (Jan 08), R3 Day 89 (Jan 10), R7 Day 93 (Jan 14), R14 Day 100 (Jan 21), R30 Day 116 (Feb 06), R60 Day 146 (Mar 08)

Day 87 — Week 13 — Friday, January 08, 2027

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Mock coding round 2

Learning objective: Understand the key invariant or decision rule for mock coding round 2; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study mock coding round 2; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Search a 2D Matrix cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 27 R60: Maximum Depth of Binary Tree; Day 57 R30: Binary Tree Maximum Path Sum; Day 73 R14: Unique Paths; Day 80 R7: Longest Increasing Subsequence; Day 84 R3: Network Delay Time; Day 86 R1: Single Number

Secondary task: Test disconnected graphs, cycles, duplicate edges, empty input, and unreachable nodes.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Search a 2D Matrix* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 88 (Jan 09), R3 Day 90 (Jan 11), R7 Day 94 (Jan 15), R14 Day 101 (Jan 22), R30 Day 117 (Feb 07), R60 Day 147 (Mar 09)

Day 88 — Week 13 — Saturday, January 09, 2027

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Weak-pattern remediation

Learning objective: Understand the key invariant or decision rule for weak-pattern remediation; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study weak-pattern remediation; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Counting Bits cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 28 R60: Longest Consecutive Sequence; Day 58 R30: Find the Duplicate Number; Day 74 R14: Clone Graph; Day 81 R7: Word Break; Day 85 R3: Find the Duplicate Number; Day 87 R1: Search a 2D Matrix

Secondary task: Compare brute force to optimized complexity and justify the trade-off.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Counting Bits* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 89 (Jan 10), R3 Day 91 (Jan 12), R7 Day 95 (Jan 16), R14 Day 102 (Jan 23), R30 Day 118 (Feb 08), R60 Day 148 (Mar 10)

Day 89 — Week 13 — Sunday, January 10, 2027

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Mixed DSA assessment

Learning objective: Understand the key invariant or decision rule for mixed dsa assessment; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study mixed dsa assessment; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Median of Two Sorted Arrays cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 29 R60: Counting Bits; Day 59 R30: Single Number; Day 75 R14: Longest Increasing Subsequence; Day 82 R7: Find the Duplicate Number; Day 86 R3: Single Number; Day 88 R1: Counting Bits

Secondary task: Add a pattern card and one counterexample to your notes.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Median of Two Sorted Arrays* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 90 (Jan 11), R3 Day 92 (Jan 13), R7 Day 96 (Jan 17), R14 Day 103 (Jan 24), R30 Day 119 (Feb 09), R60 Day 149 (Mar 11)

Day 90 — Week 13 — Monday, January 11, 2027

Phase: Advanced DSA & Algorithms **Total:** 3 hours (180 minutes)

Today's topic: Month 3 assessment and repair plan

Learning objective: Understand the key invariant or decision rule for month 3 assessment and repair plan; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Concept study: Study month 3 assessment and repair plan; write the invariant, a small example, and when the pattern applies.

65 min — New problem practice: Attempt Find the Duplicate Number cold for at least 25 minutes; derive a baseline, optimize, then test.

25 min — Second exercise / implementation: Do the listed follow-up exercise or write tests for edge cases.

20 min — Communication + log: Explain approach aloud; record complexity, mistakes, and next review dates.

Spaced-repetition queue due today: Day 30 R60: Median of Two Sorted Arrays; Day 60 R30: Search a 2D Matrix; Day 76 R14: Word Break; Day 83 R7: Binary Search; Day 87 R3: Search a 2D Matrix; Day 89 R1: Median of Two Sorted Arrays

Secondary task: Draw the graph/DP state model before coding; label transitions clearly.

Today's deliverable: Monthly milestone: record independent solve rate, median time, CS/design confidence, project/recruiting evidence, and weakest three skills.

Completion criterion: Attempt *Find the Duplicate Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 91 (Jan 12), R3 Day 93 (Jan 14), R7 Day 97 (Jan 18), R14 Day 104 (Jan 25), R30 Day 120 (Feb 10), R60 Day 150 (Mar 12)

Monthly assessment rubric: Score DSA independence (0–5), time/complexity (0–5), Java correctness (0–5), CS/design explanation (0–5), project evidence (0–5), behavioral clarity (0–5), recruiting consistency (0–5). Pass target: at least 4/5 in the month's core area and no critical gap; if below target, allocate three extra repair blocks in the next week and keep new content lighter.

Day 91 — Week 13 — Tuesday, January 12, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: OOP pillars in Java

Learning objective: Understand the key invariant or decision rule for oop pillars in java; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering oop/lld, os, databases, networking; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: System design fundamentals and Month 4 checkpoint.

Spaced-repetition queue due today: Day 31 R60: Merge Two Sorted Lists; Day 61 R30: Course Schedule; Day 77 R14: Coin Change; Day 84 R7: Network Delay Time; Day 88 R3: Counting Bits; Day 90 R1: Find the Duplicate Number

Secondary task: Draw a small diagram (thread states, schema, request path, or system components).

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 92 (Jan 13), R3 Day 94 (Jan 15), R7 Day 98 (Jan 19), R14 Day 105 (Jan 26), R30 Day 121 (Feb 11), R60 Day 151 (Mar 13)

Week 13 checkpoint: OOP/LLD, OS, databases, networking. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 92 — Week 14 — Wednesday, January 13, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Interfaces vs abstract classes; composition

Learning objective: Understand the key invariant or decision rule for interfaces vs abstract classes; composition; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Invert Binary Tree — Easy; pattern: recursive tree transformation. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study interfaces vs abstract classes; composition; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Answer two follow-ups: failure mode, scaling limit, or alternative design.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 32 R60: Binary Search; Day 62 R30: Network Delay Time; Day 78 R14: Unique Paths; Day 85 R7: Find the Duplicate Number; Day 89 R3: Median of Two Sorted Arrays; Day 91 R1: Counting Bits

Secondary task: Answer two follow-ups: failure mode, scaling limit, or alternative design.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 93 (Jan 14), R3 Day 95 (Jan 16), R7 Day 99 (Jan 20), R14 Day 106 (Jan 27), R30 Day 122 (Feb 12), R60 Day 152 (Mar 14)

Day 93 — Week 14 — Thursday, January 14, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: SOLID principles and code smells

Learning objective: Understand the key invariant or decision rule for solid principles and code smells; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study solid principles and code smells; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Write a short Java/SQL/API example where applicable.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 33 R60: Valid Parentheses; Day 63 R30: Course Schedule; Day 79 R14: Climbing Stairs; Day 86 R7: Single Number; Day 90 R3: Find the Duplicate Number; Day 92 R1: Invert Binary Tree

Secondary task: Write a short Java/SQL/API example where applicable.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 94 (Jan 15), R3 Day 96 (Jan 17), R7 Day 100 (Jan 21), R14 Day 107 (Jan 28), R30 Day 123 (Feb 13), R60 Day 153 (Mar 15)

Day 94 — Week 14 — Friday, January 15, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Design patterns: Strategy, Factory, Observer

Learning objective: Understand the key invariant or decision rule for design patterns: strategy, factory, observer; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study design patterns: strategy, factory, observer; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Explain one real-world trade-off without memorized jargon.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 34 R60: Linked List Cycle; Day 64 R30: Network Delay Time; Day 80 R14: Longest Increasing Subsequence; Day 87 R7: Search a 2D Matrix; Day 91 R3: Counting Bits; Day 93 R1: Find the Duplicate Number

Secondary task: Explain one real-world trade-off without memorized jargon.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 95 (Jan 16), R3 Day 97 (Jan 18), R7 Day 101 (Jan 22), R14 Day 108 (Jan 29), R30 Day 124 (Feb 14), R60 Day 154 (Mar 16)

Day 95 — Week 14 — Saturday, January 16, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: UML class diagrams and LLD: parking lot

Learning objective: Understand the key invariant or decision rule for uml class diagrams and lld: parking lot; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study uml class diagrams and lld: parking lot; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Prepare a 90-second verbal answer and record what was unclear.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 35 R60: Binary Tree Level Order Traversal; Day 65 R30: Clone Graph; Day 81 R14: Word Break; Day 88 R7: Counting Bits; Day 92 R3: Invert Binary Tree; Day 94 R1: Single Number

Secondary task: Prepare a 90-second verbal answer and record what was unclear.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 96 (Jan 17), R3 Day 98 (Jan 19), R7 Day 102 (Jan 23), R14 Day 109 (Jan 30), R30 Day 125 (Feb 15), R60 Day 155 (Mar 17)

Day 96 — Week 14 — Sunday, January 17, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Processes vs threads

Learning objective: Understand the key invariant or decision rule for processes vs threads; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study processes vs threads; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Create a one-page concept note with definition, example, trade-off, and interview explanation.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 36 R60: Binary Tree Maximum Path Sum; Day 66 R30: Redundant Connection; Day 82 R14: Find the Duplicate Number; Day 89 R7: Median of Two Sorted Arrays; Day 93 R3: Find the Duplicate Number; Day 95 R1: Search a 2D Matrix

Secondary task: Create a one-page concept note with definition, example, trade-off, and interview explanation.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 97 (Jan 18), R3 Day 99 (Jan 20), R7 Day 103 (Jan 24), R14 Day 110 (Jan 31), R30 Day 126 (Feb 16), R60 Day 156 (Mar 18)

Day 97 — Week 14 — Monday, January 18, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Holiday-aware adjustment: MLK Day: regular schedule; use the day for cumulative review if classes are closed.

Today's topic: Synchronization, locks, race conditions

Learning objective: Understand the key invariant or decision rule for synchronization, locks, race conditions; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study synchronization, locks, race conditions; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Draw a small diagram (thread states, schema, request path, or system components).

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 37 R60: Linked List Cycle; Day 67 R30: Number of Islands; Day 83 R14: Binary Search; Day 90 R7: Find the Duplicate Number; Day 94 R3: Single Number; Day 96 R1: Counting Bits

Secondary task: Draw a small diagram (thread states, schema, request path, or system components).

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 98 (Jan 19), R3 Day 100 (Jan 21), R7 Day 104 (Jan 25), R14 Day 111 (Feb 01), R30 Day 127 (Feb 17), R60 Day 157 (Mar 19)

Day 98 — Week 14 — Tuesday, January 19, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: OOP + OS

Learning objective: Understand the key invariant or decision rule for weekly assessment: oop + os; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering system design fundamentals and month 4 checkpoint; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Build backend API and database integration.

Spaced-repetition queue due today: Day 38 R60: Binary Tree Maximum Path Sum; Day 68 R30: Rotting Oranges; Day 84 R14: Network Delay Time; Day 91 R7: Counting Bits; Day 95 R3: Search a 2D Matrix; Day 97 R1: Median of Two Sorted Arrays

Secondary task: Answer two follow-ups: failure mode, scaling limit, or alternative design.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 99 (Jan 20), R3 Day 101 (Jan 22), R7 Day 105 (Jan 26), R14 Day 112 (Feb 02), R30 Day 128 (Feb 18), R60 Day 158 (Mar 20)

Week 14 checkpoint: System design fundamentals and Month 4 checkpoint. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 99 — Week 15 — Wednesday, January 20, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Deadlocks: conditions and prevention

Learning objective: Understand the key invariant or decision rule for deadlocks: conditions and prevention; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study deadlocks: conditions and prevention; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Write a short Java/SQL/API example where applicable.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 39 R60: Binary Tree Level Order Traversal; Day 69 R30: Min Cost to Connect All Points; Day 85 R14: Find the Duplicate Number; Day 92 R7: Invert Binary Tree; Day 96 R3: Counting Bits; Day 98 R1: Search a 2D Matrix

Secondary task: Write a short Java/SQL/API example where applicable.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 100 (Jan 21), R3 Day 102 (Jan 23), R7 Day 106 (Jan 27), R14 Day 113 (Feb 03), R30 Day 129 (Feb 19), R60 Day 159 (Mar 21)

Day 100 — Week 15 — Thursday, January 21, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Memory management, heap/stack, GC

Learning objective: Understand the key invariant or decision rule for memory management, heap/stack, gc; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Word Search — Medium; pattern: grid DFS + backtracking. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study memory management, heap/stack, gc; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Explain one real-world trade-off without memorized jargon.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 40 R60: Single Number; Day 70 R30: Rotting Oranges; Day 86 R14: Single Number; Day 93 R7: Find the Duplicate Number; Day 97 R3: Median of Two Sorted Arrays; Day 99 R1: Counting Bits

Secondary task: Explain one real-world trade-off without memorized jargon.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 101 (Jan 22), R3 Day 103 (Jan 24), R7 Day 107 (Jan 28), R14 Day 114 (Feb 04), R30 Day 130 (Feb 20), R60 Day 160 (Mar 22)

Day 101 — Week 15 — Friday, January 22, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Virtual memory and scheduling

Learning objective: Understand the key invariant or decision rule for virtual memory and scheduling; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study virtual memory and scheduling; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Prepare a 90-second verbal answer and record what was unclear.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 41 R60: Climbing Stairs; Day 71 R30: Search a 2D Matrix; Day 87 R14: Search a 2D Matrix; Day 94 R7: Single Number; Day 98 R3: Search a 2D Matrix; Day 100 R1: Word Search

Secondary task: Prepare a 90-second verbal answer and record what was unclear.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 102 (Jan 23), R3 Day 104 (Jan 25), R7 Day 108 (Jan 29), R14 Day 115 (Feb 05), R30 Day 131 (Feb 21), R60 Day 161 (Mar 23)

Day 102 — Week 15 — Saturday, January 23, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: SQL joins and aggregations

Learning objective: Understand the key invariant or decision rule for sql joins and aggregations; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study sql joins and aggregations; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Create a one-page concept note with definition, example, trade-off, and interview explanation.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 42 R60: Maximum Depth of Binary Tree; Day 72 R30: Coin Change; Day 88 R14: Counting Bits; Day 95 R7: Search a 2D Matrix; Day 99 R3: Counting Bits; Day 101 R1: Find the Duplicate Number

Secondary task: Create a one-page concept note with definition, example, trade-off, and interview explanation.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 103 (Jan 24), R3 Day 105 (Jan 26), R7 Day 109 (Jan 30), R14 Day 116 (Feb 06), R30 Day 132 (Feb 22), R60 Day 162 (Mar 24)

Day 103 — Week 15 — Sunday, January 24, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Indexes and query plans

Learning objective: Understand the key invariant or decision rule for indexes and query plans; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study indexes and query plans; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Draw a small diagram (thread states, schema, request path, or system components).

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 43 R60: Kth Largest Element in an Array; Day 73 R30: Unique Paths; Day 89 R14: Median of Two Sorted Arrays; Day 96 R7: Counting Bits; Day 100 R3: Word Search; Day 102 R1: Single Number

Secondary task: Draw a small diagram (thread states, schema, request path, or system components).

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 104 (Jan 25), R3 Day 106 (Jan 27), R7 Day 110 (Jan 31), R14 Day 117 (Feb 07), R30 Day 133 (Feb 23), R60 Day 163 (Mar 25)

Day 104 — Week 15 — Monday, January 25, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Transactions and ACID

Learning objective: Understand the key invariant or decision rule for transactions and acid; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study transactions and acid; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Answer two follow-ups: failure mode, scaling limit, or alternative design.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 44 R60: Maximum Depth of Binary Tree; Day 74 R30: Clone Graph; Day 90 R14: Find the Duplicate Number; Day 97 R7: Median of Two Sorted Arrays; Day 101 R3: Find the Duplicate Number; Day 103 R1: Search a 2D Matrix

Secondary task: Answer two follow-ups: failure mode, scaling limit, or alternative design.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 105 (Jan 26), R3 Day 107 (Jan 28), R7 Day 111 (Feb 01), R14 Day 118 (Feb 08), R30 Day 134 (Feb 24), R60 Day 164 (Mar 26)

Day 105 — Week 15 — Tuesday, January 26, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: databases

Learning objective: Understand the key invariant or decision rule for weekly assessment: databases; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering build backend api and database integration; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Testing, Docker, CI/CD, observability.

Spaced-repetition queue due today: Day 45 R60: Kth Largest Element in an Array; Day 75 R30: Longest Increasing Subsequence; Day 91 R14: Counting Bits; Day 98 R7: Search a 2D Matrix; Day 102 R3: Single Number; Day 104 R1: Counting Bits

Secondary task: Write a short Java/SQL/API example where applicable.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 106 (Jan 27); R3 Day 108 (Jan 29); R7 Day 112 (Feb 02); R14 Day 119 (Feb 09); R30 Day 135 (Feb 25); R60 Day 165 (Mar 27)

Week 15 checkpoint: Build backend API and database integration. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 106 — Week 16 — Wednesday, January 27, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Normalization and schema design

Learning objective: Understand the key invariant or decision rule for normalization and schema design; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study normalization and schema design; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Explain one real-world trade-off without memorized jargon.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 46 R60: Subsets; Day 76 R30: Word Break; Day 92 R14: Invert Binary Tree; Day 99 R7: Counting Bits; Day 103 R3: Search a 2D Matrix; Day 105 R1: Single Number

Secondary task: Explain one real-world trade-off without memorized jargon.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 107 (Jan 28), R3 Day 109 (Jan 30), R7 Day 113 (Feb 03), R14 Day 120 (Feb 10), R30 Day 136 (Feb 26), R60 Day 166 (Mar 28)

Day 107 — Week 16 — Thursday, January 28, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Isolation levels and concurrency anomalies

Learning objective: Understand the key invariant or decision rule for isolation levels and concurrency anomalies; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study isolation levels and concurrency anomalies; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Prepare a 90-second verbal answer and record what was unclear.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 47 R60: Kth Largest Element in an Array; Day 77 R30: Coin Change; Day 93 R14: Find the Duplicate Number; Day 100 R7: Word Search; Day 104 R3: Counting Bits; Day 106 R1: Search a 2D Matrix

Secondary task: Prepare a 90-second verbal answer and record what was unclear.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 108 (Jan 29), R3 Day 110 (Jan 31), R7 Day 114 (Feb 04), R14 Day 121 (Feb 11), R30 Day 137 (Feb 27), R60 Day 167 (Mar 29)

Day 108 — Week 16 — Friday, January 29, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Relational vs NoSQL trade-offs

Learning objective: Understand the key invariant or decision rule for relational vs nosql trade-offs; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study relational vs nosql trade-offs; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Create a one-page concept note with definition, example, trade-off, and interview explanation.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 48 R60: Subsets; Day 78 R30: Unique Paths; Day 94 R14: Single Number; Day 101 R7: Find the Duplicate Number; Day 105 R3: Single Number; Day 107 R1: Counting Bits

Secondary task: Create a one-page concept note with definition, example, trade-off, and interview explanation.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 109 (Jan 30), R3 Day 111 (Feb 01), R7 Day 115 (Feb 05), R14 Day 122 (Feb 12), R30 Day 138 (Feb 28), R60 Day 168 (Mar 30)

Day 109 — Week 16 — Saturday, January 30, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: HTTP/HTTPS, DNS, TCP/UDP

Learning objective: Understand the key invariant or decision rule for http/https, dns, tcp/udp; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study http/https, dns, tcp/udp; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Draw a small diagram (thread states, schema, request path, or system components).

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 49 R60: Task Scheduler; Day 79 R30: Climbing Stairs; Day 95 R14: Search a 2D Matrix; Day 102 R7: Single Number; Day 106 R3: Search a 2D Matrix; Day 108 R1: Median of Two Sorted Arrays

Secondary task: Draw a small diagram (thread states, schema, request path, or system components).

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 110 (Jan 31), R3 Day 112 (Feb 02), R7 Day 116 (Feb 06), R14 Day 123 (Feb 13), R30 Day 139 (Mar 01), R60 Day 169 (Mar 31)

Day 110 — Week 16 — Sunday, January 31, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: REST API semantics and status codes

Learning objective: Understand the key invariant or decision rule for rest api semantics and status codes; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study rest api semantics and status codes; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Answer two follow-ups: failure mode, scaling limit, or alternative design.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 50 R60: Combination Sum; Day 80 R30: Longest Increasing Subsequence; Day 96 R14: Counting Bits; Day 103 R7: Search a 2D Matrix; Day 107 R3: Counting Bits; Day 109 R1: Find the Duplicate Number

Secondary task: Answer two follow-ups: failure mode, scaling limit, or alternative design.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 111 (Feb 01), R3 Day 113 (Feb 03), R7 Day 117 (Feb 07), R14 Day 124 (Feb 14), R30 Day 140 (Mar 02), R60 Day 170 (Apr 01)

Day 111 — Week 16 — Monday, February 01, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Authentication vs authorization

Learning objective: Understand the key invariant or decision rule for authentication vs authorization; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study authentication vs authorization; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Write a short Java/SQL/API example where applicable.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 51 R60: Task Scheduler; Day 81 R30: Word Break; Day 97 R14: Median of Two Sorted Arrays; Day 104 R7: Counting Bits; Day 108 R3: Median of Two Sorted Arrays; Day 110 R1: Single Number

Secondary task: Write a short Java/SQL/API example where applicable.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 112 (Feb 02), R3 Day 114 (Feb 04), R7 Day 118 (Feb 08), R14 Day 125 (Feb 15), R30 Day 141 (Mar 03), R60 Day 171 (Apr 02)

Day 112 — Week 16 — Tuesday, February 02, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: networking + backend

Learning objective: Understand the key invariant or decision rule for weekly assessment: networking + backend; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering testing, docker, ci/cd, observability; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Project narrative, resume, STAR stories, recruiting.

Spaced-repetition queue due today: Day 52 R60: Koko Eating Bananas; Day 82 R30: Find the Duplicate Number; Day 98 R14: Search a 2D Matrix; Day 105 R7: Single Number; Day 109 R3: Find the Duplicate Number; Day 111 R1: Search a 2D Matrix

Secondary task: Explain one real-world trade-off without memorized jargon.

Today's deliverable: Checkpoint: comprehensive cumulative assessment; compare with prior scores and write a 3-item remediation plan.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 113 (Feb 03), R3 Day 115 (Feb 05), R7 Day 119 (Feb 09), R14 Day 126 (Feb 16), R30 Day 142 (Mar 04), R60 Day 172 (Apr 03)

Week 16 checkpoint: Testing, Docker, CI/CD, observability. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 113 — Week 17 — Wednesday, February 03, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Caching strategies and invalidation

Learning objective: Understand the key invariant or decision rule for caching strategies and invalidation; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study caching strategies and invalidation; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Prepare a 90-second verbal answer and record what was unclear.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 53 R60: Search in Rotated Sorted Array; Day 83 R30: Binary Search; Day 99 R14: Counting Bits; Day 106 R7: Search a 2D Matrix; Day 110 R3: Single Number; Day 112 R1: Find the Duplicate Number

Secondary task: Prepare a 90-second verbal answer and record what was unclear.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 114 (Feb 04), R3 Day 116 (Feb 06), R7 Day 120 (Feb 10), R14 Day 127 (Feb 17), R30 Day 143 (Mar 05), R60 Day 173 (Apr 04)

Day 114 — Week 17 — Thursday, February 04, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Load balancing, queues, rate limiting

Learning objective: Understand the key invariant or decision rule for load balancing, queues, rate limiting; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Valid Parentheses — Easy; pattern: stack. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study load balancing, queues, rate limiting; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Create a one-page concept note with definition, example, trade-off, and interview explanation.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 54 R60: Merge Intervals; Day 84 R30: Network Delay Time; Day 100 R14: Word Search; Day 107 R7: Counting Bits; Day 111 R3: Search a 2D Matrix; Day 113 R1: Single Number

Secondary task: Create a one-page concept note with definition, example, trade-off, and interview explanation.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 115 (Feb 05), R3 Day 117 (Feb 07), R7 Day 121 (Feb 11), R14 Day 128 (Feb 18), R30 Day 144 (Mar 06), R60 Day 174 (Apr 05)

Day 115 — Week 17 — Friday, February 05, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: System design: requirements + capacity estimates

Learning objective: Understand the key invariant or decision rule for system design: requirements + capacity estimates; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study system design: requirements + capacity estimates; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Draw a small diagram (thread states, schema, request path, or system components).

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 55 R60: Find Minimum in Rotated Sorted Array; Day 85 R30: Find the Duplicate Number; Day 101 R14: Find the Duplicate Number; Day 108 R7: Median of Two Sorted Arrays; Day 112 R3: Find the Duplicate Number; Day 114 R1: Valid Parentheses

Secondary task: Draw a small diagram (thread states, schema, request path, or system components).

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 116 (Feb 06), R3 Day 118 (Feb 08), R7 Day 122 (Feb 12), R14 Day 129 (Feb 19), R30 Day 145 (Mar 07), R60 Day 175 (Apr 06)

Day 116 — Week 17 — Saturday, February 06, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Design URL shortener

Learning objective: Understand the key invariant or decision rule for design url shortener; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study design url shortener; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Answer two follow-ups: failure mode, scaling limit, or alternative design.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 56 R60: Word Search; Day 86 R30: Single Number; Day 102 R14: Single Number; Day 109 R7: Find the Duplicate Number; Day 113 R3: Single Number; Day 115 R1: Counting Bits

Secondary task: Answer two follow-ups: failure mode, scaling limit, or alternative design.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 117 (Feb 07), R3 Day 119 (Feb 09), R7 Day 123 (Feb 13), R14 Day 130 (Feb 20), R30 Day 146 (Mar 08), R60 Day 176 (Apr 07)

Day 117 — Week 17 — Sunday, February 07, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Replication, consistency, sharding

Learning objective: Understand the key invariant or decision rule for replication, consistency, sharding; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study replication, consistency, sharding; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Write a short Java/SQL/API example where applicable.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 57 R60: Binary Tree Maximum Path Sum; Day 87 R30: Search a 2D Matrix; Day 103 R14: Search a 2D Matrix; Day 110 R7: Single Number; Day 114 R3: Valid Parentheses; Day 116 R1: Median of Two Sorted Arrays

Secondary task: Write a short Java/SQL/API example where applicable.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 118 (Feb 08), R3 Day 120 (Feb 10), R7 Day 124 (Feb 14), R14 Day 131 (Feb 21), R30 Day 147 (Mar 09), R60 Day 177 (Apr 08)

Day 118 — Week 17 — Monday, February 08, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Observability: logs, metrics, tracing

Learning objective: Understand the key invariant or decision rule for observability: logs, metrics, tracing; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study observability: logs, metrics, tracing; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Explain one real-world trade-off without memorized jargon.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 58 R60: Find the Duplicate Number; Day 88 R30: Counting Bits; Day 104 R14: Counting Bits; Day 111 R7: Search a 2D Matrix; Day 115 R3: Counting Bits; Day 117 R1: Find the Duplicate Number

Secondary task: Explain one real-world trade-off without memorized jargon.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 119 (Feb 09), R3 Day 121 (Feb 11), R7 Day 125 (Feb 15), R14 Day 132 (Feb 22), R30 Day 148 (Mar 10), R60 Day 178 (Apr 09)

Day 119 — Week 17 — Tuesday, February 09, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Mock LLD/system design discussion

Learning objective: Understand the key invariant or decision rule for mock lld/system design discussion; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering project narrative, resume, star stories, recruiting; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Mixed practice, OA simulation, Month 5 checkpoint.

Spaced-repetition queue due today: Day 59 R60: Single Number; Day 89 R30: Median of Two Sorted Arrays; Day 105 R14: Single Number; Day 112 R7: Find the Duplicate Number; Day 116 R3: Median of Two Sorted Arrays; Day 118 R1: Single Number

Secondary task: Prepare a 90-second verbal answer and record what was unclear.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 120 (Feb 10), R3 Day 122 (Feb 12), R7 Day 126 (Feb 16), R14 Day 133 (Feb 23), R30 Day 149 (Mar 11), R60 Day 179 (Apr 10)

Week 17 checkpoint: Project narrative, resume, STAR stories, recruiting. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 120 — Week 18 — Wednesday, February 10, 2027

Phase: CS Fundamentals & System Design **Total:** 3 hours (180 minutes)

Today's topic: Month 4 assessment and repair plan

Learning objective: Understand the key invariant or decision rule for month 4 assessment and repair plan; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

40 min — Technical learning: Study month 4 assessment and repair plan; create a one-page definition/example/trade-off note.

45 min — Applied exercise: Create a one-page concept note with definition, example, trade-off, and interview explanation.

35 min — Interview explanation: Give a 2-minute explanation, then answer: why this approach, what can fail, and how it scales.

30 min — Project/recruiting + log: Update notes, resume/project artifact, or recruiting tracker.

Spaced-repetition queue due today: Day 60 R60: Search a 2D Matrix; Day 90 R30: Find the Duplicate Number; Day 106 R14: Search a 2D Matrix; Day 113 R7: Single Number; Day 117 R3: Find the Duplicate Number; Day 119 R1: Median of Two Sorted Arrays

Secondary task: Create a one-page concept note with definition, example, trade-off, and interview explanation.

Today's deliverable: Monthly milestone: record independent solve rate, median time, CS/design confidence, project/recruiting evidence, and weakest three skills.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 121 (Feb 11), R3 Day 123 (Feb 13), R7 Day 127 (Feb 17), R14 Day 134 (Feb 24), R30 Day 150 (Mar 12), R60 Day 180 (Apr 11)

Monthly assessment rubric: Score DSA independence (0–5), time/complexity (0–5), Java correctness (0–5), CS/design explanation (0–5), project evidence (0–5), behavioral clarity (0–5), recruiting consistency (0–5). Pass target: at least 4/5 in the month's core area and no critical gap; if below target, allocate three extra repair blocks in the next week and keep new content lighter.

Day 121 — Week 18 — Thursday, February 11, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Project scope, user stories, API contract

Learning objective: Understand the key invariant or decision rule for project scope, user stories, api contract; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Single Number under a timer, then explain correctness and complexity aloud.

60 min — Project build: Add a unit/integration test and document how to run it.

25 min — Recruiting / behavioral: Prepare a prioritization-under-pressure story; be specific about trade-offs. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 61 R60: Course Schedule; Day 91 R30: Counting Bits; Day 107 R14: Counting Bits; Day 114 R7: Valid Parentheses; Day 118 R3: Single Number; Day 120 R1: Find the Duplicate Number

Secondary task: Add a unit/integration test and document how to run it.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 122 (Feb 12), R3 Day 124 (Feb 14), R7 Day 128 (Feb 18), R14 Day 135 (Feb 25), R30 Day 151 (Mar 13)

Day 122 — Week 18 — Friday, February 12, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Spring Boot project skeleton and layers

Learning objective: Understand the key invariant or decision rule for spring boot project skeleton and layers; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Search a 2D Matrix under a timer, then explain correctness and complexity aloud.

60 min — Project build: Update README with setup, API examples, architecture, and known limitations.

25 min — Recruiting / behavioral: Prepare a prioritization-under-pressure story; be specific about trade-offs. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 62 R60: Network Delay Time; Day 92 R30: Invert Binary Tree; Day 108 R14: Median of Two Sorted Arrays; Day 115 R7: Counting Bits; Day 119 R3: Median of Two Sorted Arrays; Day 121 R1: Single Number

Secondary task: Update README with setup, API examples, architecture, and known limitations.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 123 (Feb 13), R3 Day 125 (Feb 15), R7 Day 129 (Feb 19), R14 Day 136 (Feb 26), R30 Day 152 (Mar 14)

Day 123 — Week 18 — Saturday, February 13, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: REST endpoints and validation

Learning objective: Understand the key invariant or decision rule for rest endpoints and validation; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Counting Bits under a timer, then explain correctness and complexity aloud.

60 min — Project build: Write one evidence-based resume bullet; do not claim unmeasured impact.

25 min — Recruiting / behavioral: Prepare an ownership/leadership story without inflating your role. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 63 R60: Course Schedule; Day 93 R30: Find the Duplicate Number; Day 109 R14: Find the Duplicate Number; Day 116 R7: Median of Two Sorted Arrays; Day 120 R3: Find the Duplicate Number; Day 122 R1: Search a 2D Matrix

Secondary task: Write one evidence-based resume bullet; do not claim unmeasured impact.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 124 (Feb 14), R3 Day 126 (Feb 16), R7 Day 130 (Feb 20), R14 Day 137 (Feb 27), R30 Day 153 (Mar 15)

Day 124 — Week 18 — Sunday, February 14, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: PostgreSQL schema and migrations

Learning objective: Understand the key invariant or decision rule for postgresql schema and migrations; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Median of Two Sorted Arrays under a timer, then explain correctness and complexity aloud.

60 min — Project build: Draft or refine a STAR story from a real experience; do not invent details.

25 min — Recruiting / behavioral: Prepare an ownership/leadership story without inflating your role. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 64 R60: Network Delay Time; Day 94 R30: Single Number; Day 110 R14: Single Number; Day 117 R7: Find the Duplicate Number; Day 121 R3: Single Number; Day 123 R1: Counting Bits

Secondary task: Draft or refine a STAR story from a real experience; do not invent details.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 125 (Feb 15), R3 Day 127 (Feb 17), R7 Day 131 (Feb 21), R14 Day 138 (Feb 28), R30 Day 154 (Mar 16)

Day 125 — Week 18 — Monday, February 15, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Holiday-aware adjustment: Presidents' Day: regular schedule with extra re-solves if classes are closed.

Today's topic: Repository/service separation

Learning objective: Understand the key invariant or decision rule for repository/service separation; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Find the Duplicate Number under a timer, then explain correctness and complexity aloud.

60 min — Project build: Review one relevant internship posting and record required skills/deadline/source.

25 min — Recruiting / behavioral: Prepare an ownership/leadership story without inflating your role. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 65 R60: Clone Graph; Day 95 R30: Search a 2D Matrix; Day 111 R14: Search a 2D Matrix; Day 118 R7: Single Number; Day 122 R3: Search a 2D Matrix; Day 124 R1: Median of Two Sorted Arrays

Secondary task: Review one relevant internship posting and record required skills/deadline/source.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 126 (Feb 16), R3 Day 128 (Feb 18), R7 Day 132 (Feb 22), R14 Day 139 (Mar 01), R30 Day 155 (Mar 17)

Day 126 — Week 18 — Tuesday, February 16, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Unit tests with JUnit/Mockito

Learning objective: Understand the key invariant or decision rule for unit tests with junit/mockito; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering mixed practice, oa simulation, month 5 checkpoint; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Timed DSA rounds and graph/DP refresh.

Spaced-repetition queue due today: Day 66 R60: Redundant Connection; Day 96 R30: Counting Bits; Day 112 R14: Find the Duplicate Number; Day 119 R7: Median of Two Sorted Arrays; Day 123 R3: Counting Bits; Day 125 R1: Find the Duplicate Number

Secondary task: Make one small commit with a descriptive message; keep scope bounded.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 127 (Feb 17), R3 Day 129 (Feb 19), R7 Day 133 (Feb 23), R14 Day 140 (Mar 02), R30 Day 156 (Mar 18)

Week 18 checkpoint: Mixed practice, OA simulation, Month 5 checkpoint. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 127 — Week 19 — Wednesday, February 17, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: project checkpoint

Learning objective: Understand the key invariant or decision rule for weekly assessment: project checkpoint; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Median of Two Sorted Arrays under a timer, then explain correctness and complexity aloud.

60 min — Project build: Add a unit/integration test and document how to run it.

25 min — Recruiting / behavioral: Draft a truthful 'Tell me about yourself' answer using student background, skills, and target role. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 67 R60: Number of Islands; Day 97 R30: Median of Two Sorted Arrays; Day 113 R14: Single Number; Day 120 R7: Find the Duplicate Number; Day 124 R3: Median of Two Sorted Arrays; Day 126 R1: Counting Bits

Secondary task: Add a unit/integration test and document how to run it.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 128 (Feb 18), R3 Day 130 (Feb 20), R7 Day 134 (Feb 24), R14 Day 141 (Mar 03), R30 Day 157 (Mar 19)

Day 128 — Week 19 — Thursday, February 18, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Integration tests and test data

Learning objective: Understand the key invariant or decision rule for integration tests and test data; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Find the Duplicate Number under a timer, then explain correctness and complexity aloud.

60 min — Project build: Update README with setup, API examples, architecture, and known limitations.

25 min — Recruiting / behavioral: Draft a truthful 'Tell me about yourself' answer using student background, skills, and target role. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 68 R60: Rotting Oranges; Day 98 R30: Search a 2D Matrix; Day 114 R14: Valid Parentheses; Day 121 R7: Single Number; Day 125 R3: Find the Duplicate Number; Day 127 R1: Median of Two Sorted Arrays

Secondary task: Update README with setup, API examples, architecture, and known limitations.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 129 (Feb 19), R3 Day 131 (Feb 21), R7 Day 135 (Feb 25), R14 Day 142 (Mar 04), R30 Day 158 (Mar 20)

Day 129 — Week 19 — Friday, February 19, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Error handling and API documentation

Learning objective: Understand the key invariant or decision rule for error handling and api documentation; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Single Number under a timer, then explain correctness and complexity aloud.

60 min — Project build: Write one evidence-based resume bullet; do not claim unmeasured impact.

25 min — Recruiting / behavioral: Write a STAR story about a real technical challenge; identify your own actions and evidence. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 69 R60: Min Cost to Connect All Points; Day 99 R30: Counting Bits; Day 115 R14: Counting Bits; Day 122 R7: Search a 2D Matrix; Day 126 R3: Counting Bits; Day 128 R1: Find the Duplicate Number

Secondary task: Write one evidence-based resume bullet; do not claim unmeasured impact.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 130 (Feb 20), R3 Day 132 (Feb 22), R7 Day 136 (Feb 26), R14 Day 143 (Mar 05), R30 Day 159 (Mar 21)

Day 130 — Week 19 — Saturday, February 20, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Authentication/security basics

Learning objective: Understand the key invariant or decision rule for authentication/security basics; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Search a 2D Matrix under a timer, then explain correctness and complexity aloud.

60 min — Project build: Draft or refine a STAR story from a real experience; do not invent details.

25 min — Recruiting / behavioral: Write a STAR story about a real technical challenge; identify your own actions and evidence. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 70 R60: Rotting Oranges; Day 100 R30: Word Search; Day 116 R14: Median of Two Sorted Arrays; Day 123 R7: Counting Bits; Day 127 R3: Median of Two Sorted Arrays; Day 129 R1: Single Number

Secondary task: Draft or refine a STAR story from a real experience; do not invent details.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 131 (Feb 21), R3 Day 133 (Feb 23), R7 Day 137 (Feb 27), R14 Day 144 (Mar 06), R30 Day 160 (Mar 22)

Day 131 — Week 19 — Sunday, February 21, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Pagination, filtering, and sorting

Learning objective: Understand the key invariant or decision rule for pagination, filtering, and sorting; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search in Rotated Sorted Array — Medium; pattern: modified binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Search in Rotated Sorted Array under a timer, then explain correctness and complexity aloud.

60 min — Project build: Review one relevant internship posting and record required skills/deadline/source.

25 min — Recruiting / behavioral: Write a STAR story about a real technical challenge; identify your own actions and evidence. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 71 R60: Search a 2D Matrix; Day 101 R30: Find the Duplicate Number; Day 117 R14: Find the Duplicate Number; Day 124 R7: Median of Two Sorted Arrays; Day 128 R3: Find the Duplicate Number; Day 130 R1: Search a 2D Matrix

Secondary task: Review one relevant internship posting and record required skills/deadline/source.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 132 (Feb 22), R3 Day 134 (Feb 24), R7 Day 138 (Feb 28), R14 Day 145 (Mar 07), R30 Day 161 (Mar 23)

Day 132 — Week 19 — Monday, February 22, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Git branching, PRs, code review

Learning objective: Understand the key invariant or decision rule for git branching, prs, code review; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Median of Two Sorted Arrays under a timer, then explain correctness and complexity aloud.

60 min — Project build: Make one small commit with a descriptive message; keep scope bounded.

25 min — Recruiting / behavioral: Prepare a disagreement/teamwork story from a real class or work experience. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 72 R60: Coin Change; Day 102 R30: Single Number; Day 118 R14: Single Number; Day 125 R7: Find the Duplicate Number; Day 129 R3: Single Number; Day 131 R1: Search in Rotated Sorted Array

Secondary task: Make one small commit with a descriptive message; keep scope bounded.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 133 (Feb 23), R3 Day 135 (Feb 25), R7 Day 139 (Mar 01), R14 Day 146 (Mar 08), R30 Day 162 (Mar 24)

Day 133 — Week 19 — Tuesday, February 23, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Dockerfile and local compose

Learning objective: Understand the key invariant or decision rule for dockerfile and local compose; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering timed dsa rounds and graph/dp refresh; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Mock technical interviews and weak-pattern repair.

Spaced-repetition queue due today: Day 73 R60: Unique Paths; Day 103 R30: Search a 2D Matrix; Day 119 R14: Median of Two Sorted Arrays; Day 126 R7: Counting Bits; Day 130 R3: Search a 2D Matrix; Day 132 R1: Median of Two Sorted Arrays

Secondary task: Add a unit/integration test and document how to run it.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 134 (Feb 24), R3 Day 136 (Feb 26), R7 Day 140 (Mar 02), R14 Day 147 (Mar 09), R30 Day 163 (Mar 25)

Week 19 checkpoint: Timed DSA rounds and graph/DP refresh. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 134 — Week 20 — Wednesday, February 24, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: deployability

Learning objective: Understand the key invariant or decision rule for weekly assessment: deployability; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Counting Bits under a timer, then explain correctness and complexity aloud.

60 min — Project build: Update README with setup, API examples, architecture, and known limitations.

25 min — Recruiting / behavioral: Prepare a disagreement/teamwork story from a real class or work experience. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 74 R60: Clone Graph; Day 104 R30: Counting Bits; Day 120 R14: Find the Duplicate Number; Day 127 R7: Median of Two Sorted Arrays; Day 131 R3: Search in Rotated Sorted Array; Day 133 R1: Search a 2D Matrix

Secondary task: Update README with setup, API examples, architecture, and known limitations.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 135 (Feb 25), R3 Day 137 (Feb 27), R7 Day 141 (Mar 03), R14 Day 148 (Mar 10), R30 Day 164 (Mar 26)

Day 135 — Week 20 — Thursday, February 25, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: CI pipeline and test automation

Learning objective: Understand the key invariant or decision rule for ci pipeline and test automation; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Median of Two Sorted Arrays under a timer, then explain correctness and complexity aloud.

60 min — Project build: Write one evidence-based resume bullet; do not claim unmeasured impact.

25 min — Recruiting / behavioral: Prepare a mistake/failure story focused on accountability and what changed afterward. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 75 R60: Longest Increasing Subsequence; Day 105 R30: Single Number; Day 121 R14: Single Number; Day 128 R7: Find the Duplicate Number; Day 132 R3: Median of Two Sorted Arrays; Day 134 R1: Counting Bits

Secondary task: Write one evidence-based resume bullet; do not claim unmeasured impact.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 136 (Feb 26), R3 Day 138 (Feb 28), R7 Day 142 (Mar 04), R14 Day 149 (Mar 11), R30 Day 165 (Mar 27)

Day 136 — Week 20 — Friday, February 26, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Logging, health checks, metrics

Learning objective: Understand the key invariant or decision rule for logging, health checks, metrics; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Find the Duplicate Number under a timer, then explain correctness and complexity aloud.

60 min — Project build: Draft or refine a STAR story from a real experience; do not invent details.

25 min — Recruiting / behavioral: Prepare a mistake/failure story focused on accountability and what changed afterward. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 76 R60: Word Break; Day 106 R30: Search a 2D Matrix; Day 122 R14: Search a 2D Matrix; Day 129 R7: Single Number; Day 133 R3: Search a 2D Matrix; Day 135 R1: Median of Two Sorted Arrays

Secondary task: Draft or refine a STAR story from a real experience; do not invent details.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 137 (Feb 27), R3 Day 139 (Mar 01), R7 Day 143 (Mar 05), R14 Day 150 (Mar 12), R30 Day 166 (Mar 28)

Day 137 — Week 20 — Saturday, February 27, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Cloud deployment design and configuration

Learning objective: Understand the key invariant or decision rule for cloud deployment design and configuration; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Single Number under a timer, then explain correctness and complexity aloud.

60 min — Project build: Review one relevant internship posting and record required skills/deadline/source.

25 min — Recruiting / behavioral: Prepare a mistake/failure story focused on accountability and what changed afterward. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 77 R60: Coin Change; Day 107 R30: Counting Bits; Day 123 R14: Counting Bits; Day 130 R7: Search a 2D Matrix; Day 134 R3: Counting Bits; Day 136 R1: Find the Duplicate Number

Secondary task: Review one relevant internship posting and record required skills/deadline/source.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 138 (Feb 28), R3 Day 140 (Mar 02), R7 Day 144 (Mar 06), R14 Day 151 (Mar 13), R30 Day 167 (Mar 29)

Day 138 — Week 20 — Sunday, February 28, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: README, architecture diagram, API examples

Learning objective: Understand the key invariant or decision rule for readme, architecture diagram, api examples; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Search a 2D Matrix under a timer, then explain correctness and complexity aloud.

60 min — Project build: Make one small commit with a descriptive message; keep scope bounded.

25 min — Recruiting / behavioral: Prepare a learning-quickly story using a real example and concrete outcome. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 78 R60: Unique Paths; Day 108 R30: Median of Two Sorted Arrays; Day 124 R14: Median of Two Sorted Arrays; Day 131 R7: Search in Rotated Sorted Array; Day 135 R3: Median of Two Sorted Arrays; Day 137 R1: Single Number

Secondary task: Make one small commit with a descriptive message; keep scope bounded.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 139 (Mar 01), R3 Day 141 (Mar 03), R7 Day 145 (Mar 07), R14 Day 152 (Mar 14), R30 Day 168 (Mar 30)

Day 139 — Week 20 — Monday, March 01, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Resume bullets: impact, scope, evidence

Learning objective: Understand the key invariant or decision rule for resume bullets: impact, scope, evidence; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Counting Bits under a timer, then explain correctness and complexity aloud.

60 min — Project build: Add a unit/integration test and document how to run it.

25 min — Recruiting / behavioral: Prepare a learning-quickly story using a real example and concrete outcome. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 79 R60: Climbing Stairs; Day 109 R30: Find the Duplicate Number; Day 125 R14: Find the Duplicate Number; Day 132 R7: Median of Two Sorted Arrays; Day 136 R3: Find the Duplicate Number; Day 138 R1: Search a 2D Matrix

Secondary task: Add a unit/integration test and document how to run it.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 140 (Mar 02), R3 Day 142 (Mar 04), R7 Day 146 (Mar 08), R14 Day 153 (Mar 15), R30 Day 169 (Mar 31)

Day 140 — Week 20 — Tuesday, March 02, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Behavioral STAR: technical challenge

Learning objective: Understand the key invariant or decision rule for behavioral star: technical challenge; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering mock technical interviews and weak-pattern repair; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Behavioral, system design and project deep dives.

Spaced-repetition queue due today: Day 80 R60: Longest Increasing Subsequence; Day 110 R30: Single Number; Day 126 R14: Counting Bits; Day 133 R7: Search a 2D Matrix; Day 137 R3: Single Number; Day 139 R1: Counting Bits

Secondary task: Update README with setup, API examples, architecture, and known limitations.

Today's deliverable: Checkpoint: comprehensive cumulative assessment; compare with prior scores and write a 3-item remediation plan.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 141 (Mar 03), R3 Day 143 (Mar 05), R7 Day 147 (Mar 09), R14 Day 154 (Mar 16), R30 Day 170 (Apr 01)

Week 20 checkpoint: Mock technical interviews and weak-pattern repair. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 141 — Week 21 — Wednesday, March 03, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: project + behavioral

Learning objective: Understand the key invariant or decision rule for weekly assessment: project + behavioral; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Search a 2D Matrix under a timer, then explain correctness and complexity aloud.

60 min — Project build: Write one evidence-based resume bullet; do not claim unmeasured impact.

25 min — Recruiting / behavioral: Prepare a prioritization-under-pressure story; be specific about trade-offs. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 81 R60: Word Break; Day 111 R30: Search a 2D Matrix; Day 127 R14: Median of Two Sorted Arrays; Day 134 R7: Counting Bits; Day 138 R3: Search a 2D Matrix; Day 140 R1: Single Number

Secondary task: Write one evidence-based resume bullet; do not claim unmeasured impact.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 142 (Mar 04), R3 Day 144 (Mar 06), R7 Day 148 (Mar 10), R14 Day 155 (Mar 17), R30 Day 171 (Apr 02)

Day 142 — Week 21 — Thursday, March 04, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Project deep-dive and trade-offs

Learning objective: Understand the key invariant or decision rule for project deep-dive and trade-offs; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Counting Bits under a timer, then explain correctness and complexity aloud.

60 min — Project build: Draft or refine a STAR story from a real experience; do not invent details.

25 min — Recruiting / behavioral: Prepare a prioritization-under-pressure story; be specific about trade-offs. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 82 R60: Find the Duplicate Number; Day 112 R30: Find the Duplicate Number; Day 128 R14: Find the Duplicate Number; Day 135 R7: Median of Two Sorted Arrays; Day 139 R3: Counting Bits; Day 141 R1: Search a 2D Matrix

Secondary task: Draft or refine a STAR story from a real experience; do not invent details.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 143 (Mar 05), R3 Day 145 (Mar 07), R7 Day 149 (Mar 11), R14 Day 156 (Mar 18), R30 Day 172 (Apr 03)

Day 143 — Week 21 — Friday, March 05, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Company research and role matching

Learning objective: Understand the key invariant or decision rule for company research and role matching; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Median of Two Sorted Arrays under a timer, then explain correctness and complexity aloud.

60 min — Project build: Review one relevant internship posting and record required skills/deadline/source.

25 min — Recruiting / behavioral: Prepare a prioritization-under-pressure story; be specific about trade-offs. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 83 R60: Binary Search; Day 113 R30: Single Number; Day 129 R14: Single Number; Day 136 R7: Find the Duplicate Number; Day 140 R3: Single Number; Day 142 R1: Counting Bits

Secondary task: Review one relevant internship posting and record required skills/deadline/source.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 144 (Mar 06), R3 Day 146 (Mar 08), R7 Day 150 (Mar 12), R14 Day 157 (Mar 19), R30 Day 173 (Apr 04)

Day 144 — Week 21 — Saturday, March 06, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Application tracker and follow-up process

Learning objective: Understand the key invariant or decision rule for application tracker and follow-up process; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Find the Duplicate Number under a timer, then explain correctness and complexity aloud.

60 min — Project build: Make one small commit with a descriptive message; keep scope bounded.

25 min — Recruiting / behavioral: Prepare an ownership/leadership story without inflating your role. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 84 R60: Network Delay Time; Day 114 R30: Valid Parentheses; Day 130 R14: Search a 2D Matrix; Day 137 R7: Single Number; Day 141 R3: Search a 2D Matrix; Day 143 R1: Median of Two Sorted Arrays

Secondary task: Make one small commit with a descriptive message; keep scope bounded.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 145 (Mar 07), R3 Day 147 (Mar 09), R7 Day 151 (Mar 13), R14 Day 158 (Mar 20), R30 Day 174 (Apr 05)

Day 145 — Week 21 — Sunday, March 07, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Online assessment simulation

Learning objective: Understand the key invariant or decision rule for online assessment simulation; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Single Number under a timer, then explain correctness and complexity aloud.

60 min — Project build: Add a unit/integration test and document how to run it.

25 min — Recruiting / behavioral: Prepare an ownership/leadership story without inflating your role. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 85 R60: Find the Duplicate Number; Day 115 R30: Counting Bits; Day 131 R14: Search in Rotated Sorted Array; Day 138 R7: Search a 2D Matrix; Day 142 R3: Counting Bits; Day 144 R1: Find the Duplicate Number

Secondary task: Add a unit/integration test and document how to run it.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 146 (Mar 08), R3 Day 148 (Mar 10), R7 Day 152 (Mar 14), R14 Day 159 (Mar 21), R30 Day 175 (Apr 06)

Day 146 — Week 21 — Monday, March 08, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Networking message and recruiter pitch

Learning objective: Understand the key invariant or decision rule for networking message and recruiter pitch; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Search a 2D Matrix under a timer, then explain correctness and complexity aloud.

60 min — Project build: Update README with setup, API examples, architecture, and known limitations.

25 min — Recruiting / behavioral: Prepare an ownership/leadership story without inflating your role. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 86 R60: Single Number; Day 116 R30: Median of Two Sorted Arrays; Day 132 R14: Median of Two Sorted Arrays; Day 139 R7: Counting Bits; Day 143 R3: Median of Two Sorted Arrays; Day 145 R1: Single Number

Secondary task: Update README with setup, API examples, architecture, and known limitations.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 147 (Mar 09), R3 Day 149 (Mar 11), R7 Day 153 (Mar 15), R14 Day 160 (Mar 22), R30 Day 176 (Apr 07)

Day 147 — Week 21 — Tuesday, March 09, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Mock behavioral interview

Learning objective: Understand the key invariant or decision rule for mock behavioral interview; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering behavioral, system design and project deep dives; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Company-style assessments and final revisions.

Spaced-repetition queue due today: Day 87 R60: Search a 2D Matrix; Day 117 R30: Find the Duplicate Number; Day 133 R14: Search a 2D Matrix; Day 140 R7: Single Number; Day 144 R3: Find the Duplicate Number; Day 146 R1: Search a 2D Matrix

Secondary task: Write one evidence-based resume bullet; do not claim unmeasured impact.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 148 (Mar 10); R3 Day 150 (Mar 12); R7 Day 154 (Mar 16); R14 Day 161 (Mar 23); R30 Day 177 (Apr 08)

Week 21 checkpoint: Behavioral, system design and project deep dives. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 148 — Week 22 — Wednesday, March 10, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Resume/GitHub audit

Learning objective: Understand the key invariant or decision rule for resume/github audit; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Single Number under a timer, then explain correctness and complexity aloud.

60 min — Project build: Draft or refine a STAR story from a real experience; do not invent details.

25 min — Recruiting / behavioral: Draft a truthful 'Tell me about yourself' answer using student background, skills, and target role. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 88 R60: Counting Bits; Day 118 R30: Single Number; Day 134 R14: Counting Bits; Day 141 R7: Search a 2D Matrix; Day 145 R3: Single Number; Day 147 R1: Find the Duplicate Number

Secondary task: Draft or refine a STAR story from a real experience; do not invent details.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 149 (Mar 11), R3 Day 151 (Mar 13), R7 Day 155 (Mar 17), R14 Day 162 (Mar 24), R30 Day 178 (Apr 09)

Day 149 — Week 22 — Thursday, March 11, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Mixed DSA + project integration

Learning objective: Understand the key invariant or decision rule for mixed dsa + project integration; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Search a 2D Matrix under a timer, then explain correctness and complexity aloud.

60 min — Project build: Review one relevant internship posting and record required skills/deadline/source.

25 min — Recruiting / behavioral: Draft a truthful 'Tell me about yourself' answer using student background, skills, and target role. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 89 R60: Median of Two Sorted Arrays; Day 119 R30: Median of Two Sorted Arrays; Day 135 R14: Median of Two Sorted Arrays; Day 142 R7: Counting Bits; Day 146 R3: Search a 2D Matrix; Day 148 R1: Single Number

Secondary task: Review one relevant internship posting and record required skills/deadline/source.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 150 (Mar 12), R3 Day 152 (Mar 14), R7 Day 156 (Mar 18), R14 Day 163 (Mar 25), R30 Day 179 (Apr 10)

Day 150 — Week 22 — Friday, March 12, 2027

Phase: Project, Mixed Practice & Recruiting **Total:** 3 hours (180 minutes)

Today's topic: Month 5 assessment and application review

Learning objective: Understand the key invariant or decision rule for month 5 assessment and application review; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

45 min — DSA practice: Do Counting Bits under a timer, then explain correctness and complexity aloud.

60 min — Project build: Make one small commit with a descriptive message; keep scope bounded.

25 min — Recruiting / behavioral: Write a STAR story about a real technical challenge; identify your own actions and evidence. Review one suitable posting and log its actual deadline/source.

20 min — Notes and review scheduling: Commit code or notes; log evidence and next review dates.

Spaced-repetition queue due today: Day 90 R60: Find the Duplicate Number; Day 120 R30: Find the Duplicate Number; Day 136 R14: Find the Duplicate Number; Day 143 R7: Median of Two Sorted Arrays; Day 147 R3: Find the Duplicate Number; Day 149 R1: Search a 2D Matrix

Secondary task: Make one small commit with a descriptive message; keep scope bounded.

Today's deliverable: Monthly milestone: record independent solve rate, median time, CS/design confidence, project/recruiting evidence, and weakest three skills.

Completion criterion: Produce a saved artifact (diagram, test, API, README, resume edit, or STAR draft) and explain one trade-off.

Anchor review dates: R0 today; R1 Day 151 (Mar 13), R3 Day 153 (Mar 15), R7 Day 157 (Mar 19), R14 Day 164 (Mar 26), R30 Day 180 (Apr 11)

Monthly assessment rubric: Score DSA independence (0–5), time/complexity (0–5), Java correctness (0–5), CS/design explanation (0–5), project evidence (0–5), behavioral clarity (0–5), recruiting consistency (0–5). Pass target: at least 4/5 in the month's core area and no critical gap; if below target, allocate three extra repair blocks in the next week and keep new content lighter.

Day 151 — Week 22 — Saturday, March 13, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Timed arrays/hashing round

Learning objective: Understand the key invariant or decision rule for timed arrays/hashing round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Encode and Decode Strings — Medium; pattern: serialization boundaries. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Encode and Decode Strings under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Score communication, correctness, complexity, edge cases, and time management.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 91 R60: Counting Bits; Day 121 R30: Single Number; Day 137 R14: Single Number; Day 144 R7: Find the Duplicate Number; Day 148 R3: Single Number; Day 150 R1: Counting Bits

Secondary task: Score communication, correctness, complexity, edge cases, and time management.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Encode and Decode Strings* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 152 (Mar 14), R3 Day 154 (Mar 16), R7 Day 158 (Mar 20), R14 Day 165 (Mar 27)

Day 152 — Week 22 — Sunday, March 14, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Timed two pointers/sliding window round

Learning objective: Understand the key invariant or decision rule for timed two pointers/sliding window round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Merge Two Sorted Lists — Easy; pattern: dummy node. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Merge Two Sorted Lists under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Practice a 2-minute project explanation followed by five technical follow-ups.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 92 R60: Invert Binary Tree; Day 122 R30: Search a 2D Matrix; Day 138 R14: Search a 2D Matrix; Day 145 R7: Single Number; Day 149 R3: Search a 2D Matrix; Day 151 R1: Encode and Decode Strings

Secondary task: Practice a 2-minute project explanation followed by five technical follow-ups.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Merge Two Sorted Lists* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 153 (Mar 15), R3 Day 155 (Mar 17), R7 Day 159 (Mar 21), R14 Day 166 (Mar 28)

Day 153 — Week 22 — Monday, March 15, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Holiday-aware adjustment: Spring-break period may vary by UNT calendar; use the lighter review/project maintenance format if on break.

Today's topic: Timed binary search round

Learning objective: Understand the key invariant or decision rule for timed binary search round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Binary Search — Easy; pattern: search invariant. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Binary Search under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Run a recovery drill: identify the stuck point, simplify, and test a smaller case.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 93 R60: Find the Duplicate Number; Day 123 R30: Counting Bits; Day 139 R14: Counting Bits; Day 146 R7: Search a 2D Matrix; Day 150 R3: Counting Bits; Day 152 R1: Merge Two Sorted Lists

Secondary task: Run a recovery drill: identify the stuck point, simplify, and test a smaller case.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Binary Search* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 154 (Mar 16), R3 Day 156 (Mar 18), R7 Day 160 (Mar 22), R14 Day 167 (Mar 29)

Day 154 — Week 22 — Tuesday, March 16, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Timed linked list/stack round

Learning objective: Understand the key invariant or decision rule for timed linked list/stack round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Longest Common Subsequence — Medium; pattern: 2D DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering company-style assessments and final revisions; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Final readiness, confidence, sleep and execution.

Spaced-repetition queue due today: Day 94 R60: Single Number; Day 124 R30: Median of Two Sorted Arrays; Day 140 R14: Single Number; Day 147 R7: Find the Duplicate Number; Day 151 R3: Encode and Decode Strings; Day 153 R1: Binary Search

Secondary task: Do a 10-minute behavioral answer, then cut it to a clear 90-second STAR response.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 155 (Mar 17), R3 Day 157 (Mar 19), R7 Day 161 (Mar 23), R14 Day 168 (Mar 30)

Week 22 checkpoint: Company-style assessments and final revisions. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 155 — Week 23 — Wednesday, March 17, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Timed tree DFS/BFS round

Learning objective: Understand the key invariant or decision rule for timed tree dfs/bfs round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Invert Binary Tree — Easy; pattern: recursive tree transformation. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Invert Binary Tree under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Update the weak-pattern list and choose one repair action.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 95 R60: Search a 2D Matrix; Day 125 R30: Find the Duplicate Number; Day 141 R14: Search a 2D Matrix; Day 148 R7: Single Number; Day 152 R3: Merge Two Sorted Lists; Day 154 R1: Longest Common Subsequence

Secondary task: Update the weak-pattern list and choose one repair action.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Invert Binary Tree* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 156 (Mar 18), R3 Day 158 (Mar 20), R7 Day 162 (Mar 24), R14 Day 169 (Mar 31)

Day 156 — Week 23 — Thursday, March 18, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Timed backtracking/heap round

Learning objective: Understand the key invariant or decision rule for timed backtracking/heap round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Task Scheduler — Medium; pattern: greedy counting. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Task Scheduler under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Use interview narration: clarify, baseline, optimize, prove, analyze, test.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 96 R60: Counting Bits; Day 126 R30: Counting Bits; Day 142 R14: Counting Bits; Day 149 R7: Search a 2D Matrix; Day 153 R3: Binary Search; Day 155 R1: Invert Binary Tree

Secondary task: Use interview narration: clarify, baseline, optimize, prove, analyze, test.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Task Scheduler* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 157 (Mar 19), R3 Day 159 (Mar 21), R7 Day 163 (Mar 25), R14 Day 170 (Apr 01)

Day 157 — Week 23 — Friday, March 19, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: mixed DSA

Learning objective: Understand the key invariant or decision rule for weekly assessment: mixed dsa; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Search a 2D Matrix under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Score communication, correctness, complexity, edge cases, and time management.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 97 R60: Median of Two Sorted Arrays; Day 127 R30: Median of Two Sorted Arrays; Day 143 R14: Median of Two Sorted Arrays; Day 150 R7: Counting Bits; Day 154 R3: Longest Common Subsequence; Day 156 R1: Task Scheduler

Secondary task: Score communication, correctness, complexity, edge cases, and time management.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Search a 2D Matrix* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 158 (Mar 20), R3 Day 160 (Mar 22), R7 Day 164 (Mar 26), R14 Day 171 (Apr 02)

Day 158 — Week 23 — Saturday, March 20, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Timed graph traversal round

Learning objective: Understand the key invariant or decision rule for timed graph traversal round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Min Cost to Connect All Points — Medium; pattern: minimum spanning tree. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Min Cost to Connect All Points under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Practice a 2-minute project explanation followed by five technical follow-ups.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 98 R60: Search a 2D Matrix; Day 128 R30: Find the Duplicate Number; Day 144 R14: Find the Duplicate Number; Day 151 R7: Encode and Decode Strings; Day 155 R3: Invert Binary Tree; Day 157 R1: Search a 2D Matrix

Secondary task: Practice a 2-minute project explanation followed by five technical follow-ups.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Min Cost to Connect All Points* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 159 (Mar 21), R3 Day 161 (Mar 23), R7 Day 165 (Mar 27), R14 Day 172 (Apr 03)

Day 159 — Week 23 — Sunday, March 21, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Timed topological sort/Union-Find round

Learning objective: Understand the key invariant or decision rule for timed topological sort/union-find round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Course Schedule — Medium; pattern: cycle detection / topological sort. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Course Schedule under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Run a recovery drill: identify the stuck point, simplify, and test a smaller case.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 99 R60: Counting Bits; Day 129 R30: Single Number; Day 145 R14: Single Number; Day 152 R7: Merge Two Sorted Lists; Day 156 R3: Task Scheduler; Day 158 R1: Min Cost to Connect All Points

Secondary task: Run a recovery drill: identify the stuck point, simplify, and test a smaller case.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Course Schedule* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 160 (Mar 22), R3 Day 162 (Mar 24), R7 Day 166 (Mar 28), R14 Day 173 (Apr 04)

Day 160 — Week 23 — Monday, March 22, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Timed shortest path round

Learning objective: Understand the key invariant or decision rule for timed shortest path round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Network Delay Time — Medium; pattern: Dijkstra. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Network Delay Time under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Do a 10-minute behavioral answer, then cut it to a clear 90-second STAR response.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 100 R60: Word Search; Day 130 R30: Search a 2D Matrix; Day 146 R14: Search a 2D Matrix; Day 153 R7: Binary Search; Day 157 R3: Search a 2D Matrix; Day 159 R1: Course Schedule

Secondary task: Do a 10-minute behavioral answer, then cut it to a clear 90-second STAR response.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Network Delay Time* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 161 (Mar 23), R3 Day 163 (Mar 25), R7 Day 167 (Mar 29), R14 Day 174 (Apr 05)

Day 161 — Week 23 — Tuesday, March 23, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Timed 1D DP round

Learning objective: Understand the key invariant or decision rule for timed 1d dp round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Coin Change — Medium; pattern: unbounded knapsack. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering final readiness, confidence, sleep and execution; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Consolidate prior patterns; test independently; repair the weakest skill.

Spaced-repetition queue due today: Day 101 R60: Find the Duplicate Number; Day 131 R30: Search in Rotated Sorted Array; Day 147 R14: Find the Duplicate Number; Day 154 R7: Longest Common Subsequence; Day 158 R3: Min Cost to Connect All Points; Day 160 R1: Network Delay Time

Secondary task: Update the weak-pattern list and choose one repair action.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 162 (Mar 24), R3 Day 164 (Mar 26), R7 Day 168 (Mar 30), R14 Day 175 (Apr 06)

Week 23 checkpoint: Final readiness, confidence, sleep and execution. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 162 — Week 24 — Wednesday, March 24, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Timed 2D DP round

Learning objective: Understand the key invariant or decision rule for timed 2d dp round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Unique Paths — Medium; pattern: grid DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Unique Paths under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Use interview narration: clarify, baseline, optimize, prove, analyze, test.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 102 R60: Single Number; Day 132 R30: Median of Two Sorted Arrays; Day 148 R14: Single Number; Day 155 R7: Invert Binary Tree; Day 159 R3: Course Schedule; Day 161 R1: Coin Change

Secondary task: Use interview narration: clarify, baseline, optimize, prove, analyze, test.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Unique Paths* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 163 (Mar 25), R3 Day 165 (Mar 27), R7 Day 169 (Mar 31), R14 Day 176 (Apr 07)

Day 163 — Week 24 — Thursday, March 25, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Timed bit/mixed hard round

Learning objective: Understand the key invariant or decision rule for timed bit/mixed hard round; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Find the Duplicate Number under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Score communication, correctness, complexity, edge cases, and time management.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 103 R60: Search a 2D Matrix; Day 133 R30: Search a 2D Matrix; Day 149 R14: Search a 2D Matrix; Day 156 R7: Task Scheduler; Day 160 R3: Network Delay Time; Day 162 R1: Unique Paths

Secondary task: Score communication, correctness, complexity, edge cases, and time management.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Find the Duplicate Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 164 (Mar 26), R3 Day 166 (Mar 28), R7 Day 170 (Apr 01), R14 Day 177 (Apr 08)

Day 164 — Week 24 — Friday, March 26, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: algorithms

Learning objective: Understand the key invariant or decision rule for weekly assessment: algorithms; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Single Number under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Practice a 2-minute project explanation followed by five technical follow-ups.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 104 R60: Counting Bits; Day 134 R30: Counting Bits; Day 150 R14: Counting Bits; Day 157 R7: Search a 2D Matrix; Day 161 R3: Coin Change; Day 163 R1: Find the Duplicate Number

Secondary task: Practice a 2-minute project explanation followed by five technical follow-ups.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Single Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 165 (Mar 27), R3 Day 167 (Mar 29), R7 Day 171 (Apr 02), R14 Day 178 (Apr 09)

Day 165 — Week 24 — Saturday, March 27, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Mock technical interview 1

Learning objective: Understand the key invariant or decision rule for mock technical interview 1; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Search a 2D Matrix under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Run a recovery drill: identify the stuck point, simplify, and test a smaller case.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 105 R60: Single Number; Day 135 R30: Median of Two Sorted Arrays; Day 151 R14: Encode and Decode Strings; Day 158 R7: Min Cost to Connect All Points; Day 162 R3: Unique Paths; Day 164 R1: Single Number

Secondary task: Run a recovery drill: identify the stuck point, simplify, and test a smaller case.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Search a 2D Matrix* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 166 (Mar 28), R3 Day 168 (Mar 30), R7 Day 172 (Apr 03), R14 Day 179 (Apr 10)

Day 166 — Week 24 — Sunday, March 28, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Mock technical interview 2

Learning objective: Understand the key invariant or decision rule for mock technical interview 2; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Counting Bits under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Do a 10-minute behavioral answer, then cut it to a clear 90-second STAR response.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 106 R60: Search a 2D Matrix; Day 136 R30: Find the Duplicate Number; Day 152 R14: Merge Two Sorted Lists; Day 159 R7: Course Schedule; Day 163 R3: Find the Duplicate Number; Day 165 R1: Search a 2D Matrix

Secondary task: Do a 10-minute behavioral answer, then cut it to a clear 90-second STAR response.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Counting Bits* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 167 (Mar 29), R3 Day 169 (Mar 31), R7 Day 173 (Apr 04), R14 Day 180 (Apr 11)

Day 167 — Week 24 — Monday, March 29, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Behavioral mock: ownership/failure

Learning objective: Understand the key invariant or decision rule for behavioral mock: ownership/failure; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Median of Two Sorted Arrays under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Update the weak-pattern list and choose one repair action.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 107 R60: Counting Bits; Day 137 R30: Single Number; Day 153 R14: Binary Search; Day 160 R7: Network Delay Time; Day 164 R3: Single Number; Day 166 R1: Counting Bits

Secondary task: Update the weak-pattern list and choose one repair action.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Median of Two Sorted Arrays* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 168 (Mar 30), R3 Day 170 (Apr 01), R7 Day 174 (Apr 05)

Day 168 — Week 24 — Tuesday, March 30, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: System design mock: URL shortener

Learning objective: Understand the key invariant or decision rule for system design mock: url shortener; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering consolidate prior patterns; test independently; repair the weakest skill; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Consolidate prior patterns; test independently; repair the weakest skill.

Spaced-repetition queue due today: Day 108 R60: Median of Two Sorted Arrays; Day 138 R30: Search a 2D Matrix; Day 154 R14: Longest Common Subsequence; Day 161 R7: Coin Change; Day 165 R3: Search a 2D Matrix; Day 167 R1: Median of Two Sorted Arrays

Secondary task: Use interview narration: clarify, baseline, optimize, prove, analyze, test.

Today's deliverable: Weekly assessment: timed problem(s), verbal explanation, error classification, and next-week adjustment.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 169 (Mar 31), R3 Day 171 (Apr 02), R7 Day 175 (Apr 06)

Week 24 checkpoint: Consolidate prior patterns; test independently; repair the weakest skill. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 169 — Week 25 — Wednesday, March 31, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: LLD mock: parking lot/library

Learning objective: Understand the key invariant or decision rule for lld mock: parking lot/library; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Counting Bits under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Score communication, correctness, complexity, edge cases, and time management.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 109 R60: Find the Duplicate Number; Day 139 R30: Counting Bits; Day 155 R14: Invert Binary Tree; Day 162 R7: Unique Paths; Day 166 R3: Counting Bits; Day 168 R1: Search a 2D Matrix

Secondary task: Score communication, correctness, complexity, edge cases, and time management.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Counting Bits* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 170 (Apr 01), R3 Day 172 (Apr 03), R7 Day 176 (Apr 07)

Day 170 — Week 25 — Thursday, April 01, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Project deep-dive mock

Learning objective: Understand the key invariant or decision rule for project deep-dive mock; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Median of Two Sorted Arrays under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Practice a 2-minute project explanation followed by five technical follow-ups.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 110 R60: Single Number; Day 140 R30: Single Number; Day 156 R14: Task Scheduler; Day 163 R7: Find the Duplicate Number; Day 167 R3: Median of Two Sorted Arrays; Day 169 R1: Counting Bits

Secondary task: Practice a 2-minute project explanation followed by five technical follow-ups.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Median of Two Sorted Arrays* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 171 (Apr 02), R3 Day 173 (Apr 04), R7 Day 177 (Apr 08)

Day 171 — Week 25 — Friday, April 02, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Weekly assessment: interview simulation

Learning objective: Understand the key invariant or decision rule for weekly assessment: interview simulation; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Find the Duplicate Number under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Run a recovery drill: identify the stuck point, simplify, and test a smaller case.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 111 R60: Search a 2D Matrix; Day 141 R30: Search a 2D Matrix; Day 157 R14: Search a 2D Matrix; Day 164 R7: Single Number; Day 168 R3: Search a 2D Matrix; Day 170 R1: Median of Two Sorted Arrays

Secondary task: Run a recovery drill: identify the stuck point, simplify, and test a smaller case.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Find the Duplicate Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 172 (Apr 03), R3 Day 174 (Apr 05), R7 Day 178 (Apr 09)

Day 172 — Week 25 — Saturday, April 03, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Weakest two patterns: targeted repair

Learning objective: Understand the key invariant or decision rule for weakest two patterns: targeted repair; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Single Number under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Do a 10-minute behavioral answer, then cut it to a clear 90-second STAR response.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 112 R60: Find the Duplicate Number; Day 142 R30: Counting Bits; Day 158 R14: Min Cost to Connect All Points; Day 165 R7: Search a 2D Matrix; Day 169 R3: Counting Bits; Day 171 R1: Find the Duplicate Number

Secondary task: Do a 10-minute behavioral answer, then cut it to a clear 90-second STAR response.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Single Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 173 (Apr 04), R3 Day 175 (Apr 06), R7 Day 179 (Apr 10)

Day 173 — Week 25 — Sunday, April 04, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Re-solve previously missed problems set A

Learning objective: Understand the key invariant or decision rule for re-solve previously missed problems set a; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Search a 2D Matrix under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Update the weak-pattern list and choose one repair action.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 113 R60: Single Number; Day 143 R30: Median of Two Sorted Arrays; Day 159 R14: Course Schedule; Day 166 R7: Counting Bits; Day 170 R3: Median of Two Sorted Arrays; Day 172 R1: Single Number

Secondary task: Update the weak-pattern list and choose one repair action.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Search a 2D Matrix* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 174 (Apr 05), R3 Day 176 (Apr 07), R7 Day 180 (Apr 11)

Day 174 — Week 25 — Monday, April 05, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Re-solve previously missed problems set B

Learning objective: Understand the key invariant or decision rule for re-solve previously missed problems set b; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Counting Bits under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Use interview narration: clarify, baseline, optimize, prove, analyze, test.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 114 R60: Valid Parentheses; Day 144 R30: Find the Duplicate Number; Day 160 R14: Network Delay Time; Day 167 R7: Median of Two Sorted Arrays; Day 171 R3: Find the Duplicate Number; Day 173 R1: Search a 2D Matrix

Secondary task: Use interview narration: clarify, baseline, optimize, prove, analyze, test.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Counting Bits* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 175 (Apr 06), R3 Day 177 (Apr 08)

Day 175 — Week 25 — Tuesday, April 06, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Company-style OA simulation

Learning objective: Understand the key invariant or decision rule for company-style oa simulation; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

30 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed assessment: Use a weekly mixed set covering consolidate prior patterns; test independently; repair the weakest skill; no editorials during the timer.

35 min — Review errors: Tag each miss by pattern, reasoning, Java implementation, edge case, or communication; choose one fix.

35 min — Re-solve: Re-solve the hardest miss from memory, then test a boundary case and state complexity.

20 min — Weekly summary: Summarize this week's patterns; next week's focus: Consolidate prior patterns; test independently; repair the weakest skill.

Spaced-repetition queue due today: Day 115 R60: Counting Bits; Day 145 R30: Single Number; Day 161 R14: Coin Change; Day 168 R7: Search a 2D Matrix; Day 172 R3: Single Number; Day 174 R1: Counting Bits

Secondary task: Score communication, correctness, complexity, edge cases, and time management.

Today's deliverable: Checkpoint: comprehensive cumulative assessment; compare with prior scores and write a 3-item remediation plan.

Completion criterion: Complete the timed set, review every miss, and identify one specific change for next week.

Anchor review dates: R0 today; R1 Day 176 (Apr 07), R3 Day 178 (Apr 09)

Week 25 checkpoint: Consolidate prior patterns; test independently; repair the weakest skill. Adjust next week by giving the weakest category three extra 20-minute repair blocks.

Day 176 — Week 26 — Wednesday, April 07, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Behavioral mock: teamwork/feedback

Learning objective: Understand the key invariant or decision rule for behavioral mock: teamwork/feedback; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Search a 2D Matrix — Medium; pattern: binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Search a 2D Matrix under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Practice a 2-minute project explanation followed by five technical follow-ups.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 116 R60: Median of Two Sorted Arrays; Day 146 R30: Search a 2D Matrix; Day 162 R14: Unique Paths; Day 169 R7: Counting Bits; Day 173 R3: Search a 2D Matrix; Day 175 R1: Single Number

Secondary task: Practice a 2-minute project explanation followed by five technical follow-ups.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Search a 2D Matrix* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 177 (Apr 08), R3 Day 179 (Apr 10)

Day 177 — Week 26 — Thursday, April 08, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Resume walkthrough and recruiter pitch

Learning objective: Understand the key invariant or decision rule for resume walkthrough and recruiter pitch; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Counting Bits — Easy; pattern: bit DP. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Counting Bits under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Run a recovery drill: identify the stuck point, simplify, and test a smaller case.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 117 R60: Find the Duplicate Number; Day 147 R30: Find the Duplicate Number; Day 163 R14: Find the Duplicate Number; Day 170 R7: Median of Two Sorted Arrays; Day 174 R3: Counting Bits; Day 176 R1: Search a 2D Matrix

Secondary task: Run a recovery drill: identify the stuck point, simplify, and test a smaller case.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Counting Bits* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 178 (Apr 09), R3 Day 180 (Apr 11)

Day 178 — Week 26 — Friday, April 09, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Final system design review

Learning objective: Understand the key invariant or decision rule for final system design review; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Median of Two Sorted Arrays — Hard; pattern: partition binary search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Median of Two Sorted Arrays under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Do a 10-minute behavioral answer, then cut it to a clear 90-second STAR response.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 118 R60: Single Number; Day 148 R30: Single Number; Day 164 R14: Single Number; Day 171 R7: Find the Duplicate Number; Day 175 R3: Single Number; Day 177 R1: Counting Bits

Secondary task: Do a 10-minute behavioral answer, then cut it to a clear 90-second STAR response.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Median of Two Sorted Arrays* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 179 (Apr 10)

Day 179 — Week 26 — Saturday, April 10, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Final DSA mixed assessment

Learning objective: Understand the key invariant or decision rule for final dsa mixed assessment; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Find the Duplicate Number — Medium; pattern: cycle detection / value search. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Find the Duplicate Number under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Update the weak-pattern list and choose one repair action.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 119 R60: Median of Two Sorted Arrays; Day 149 R30: Search a 2D Matrix; Day 165 R14: Search a 2D Matrix; Day 172 R7: Single Number; Day 176 R3: Search a 2D Matrix; Day 178 R1: Median of Two Sorted Arrays

Secondary task: Update the weak-pattern list and choose one repair action.

Today's deliverable: Exit check: explain today's pattern without notes, state complexity, test two edge cases, and mark Strong / Needs Practice / Relearn.

Completion criterion: Attempt *Find the Duplicate Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; R1 Day 180 (Apr 11)

Day 180 — Week 26 — Sunday, April 11, 2027

Phase: Interview Simulation & Final Preparation **Total:** 3 hours (180 minutes)

Today's topic: Final readiness assessment and next-30-day plan

Learning objective: Understand the key invariant or decision rule for final readiness assessment and next-30-day plan; implement or explain it in Java where applicable; test boundary cases; communicate complexity and trade-offs.

Today's anchor problem: Single Number — Easy; pattern: XOR properties. Treat as a new attempt unless this title has already appeared in your log; if previously solved, use it as a cold re-solve and select a new unseen problem from the same bank.

Three-hour plan

35 min — Spaced review: Use the exact due queue below. Recall first; re-solve without notes; log Strong / Needs Practice / Relearn.

60 min — Timed interview round: Solve Single Number under interview conditions; narrate clarification, baseline, optimization, tests.

35 min — CS/design/behavioral: Use interview narration: clarify, baseline, optimize, prove, analyze, test.

30 min — Error repair: Repair one high-impact mistake and rerun tests; if no coding bug, rehearse the weakest explanation.

20 min — Readiness log: Score correctness, independence, communication, complexity, edge cases and time; set tomorrow's top repair.

Spaced-repetition queue due today: Day 120 R60: Find the Duplicate Number; Day 150 R30: Counting Bits; Day 166 R14: Counting Bits; Day 173 R7: Search a 2D Matrix; Day 177 R3: Counting Bits; Day 179 R1: Find the Duplicate Number

Secondary task: Use interview narration: clarify, baseline, optimize, prove, analyze, test.

Today's deliverable: Monthly milestone: record independent solve rate, median time, CS/design confidence, project/recruiting evidence, and weakest three skills.

Completion criterion: Attempt *Single Number* independently, explain the pattern and complexity, and log the result; if stuck, record the smallest failing case.

Anchor review dates: R0 today; no future intervals fall within this 180-day plan; continue reviewing after Day 180 as needed

Monthly assessment rubric: Score DSA independence (0–5), time/complexity (0–5), Java correctness (0–5), CS/design explanation (0–5), project evidence (0–5), behavioral clarity (0–5), recruiting consistency (0–5). Pass target: at least 4/5 in the month's core area and no critical gap; if below target, allocate three extra repair blocks in the next week and keep new content lighter.

5. Beginner-Friendly Backend Project Plan

Chosen project: Campus Task Tracker API. Build this from your own understanding, starting with a small working version. It lets a user create, list, update, complete, and delete tasks. Use Java and Spring Boot only after you can explain the Java code; begin with an in-memory list if databases are new, then add a database as a later milestone. Do not add AI features just to make the project sound advanced.

Scope ladder — finish one level before moving on

Stage	Build	Proof you understand it
1. Java basics	Task class; title, description, status, due date; add/list/complete tasks in a console app.	Explain fields, constructors, methods, collections, and validation.
2. REST API	Create GET/POST/PATCH/DELETE endpoints with Spring Boot; validate required fields and return sensible status codes.	Demo each endpoint with curl or Postman; explain request/response JSON.
3. Persistence	Replace in-memory storage with a relational database such as PostgreSQL or H2; keep controller/service/repository responsibilities clear.	Explain schema, primary key, CRUD queries, and what changes when the app restarts.
4. Quality	Add tests for normal cases, missing titles, invalid IDs, and status changes; document how to run them.	Run tests from a clean checkout and explain what is not tested.
5. Packaging	Optional: Docker, a basic CI workflow, health endpoint, and deployment only after the app and tests work locally.	Explain each configuration file; omit any tool you cannot explain.

Suggested timeline

• **Days 121–124:** Define task fields, API endpoints, examples, and README scope.

• **Days 125–129:** Implement the Task model and in-memory CRUD endpoints; keep the design simple.

• **Days 130–134:** Add database persistence and explain each layer.

• **Days 135–139:** Add unit/integration tests and validation/error responses.

• **Days 140–144:** Optional Docker/CI/health check, only if the previous stages are stable.

• **Days 145–149:** Finish README, sample requests, architecture sketch, limitations, and a short demo.

• **Days 150–180:** Practice a truthful project walkthrough and make small fixes based on mock interview questions.

Project interview questions

• What problem does this solve, and what did you intentionally leave out?

• Walk through one request from controller to service to repository/database.

• How do you handle invalid input, missing IDs, and unexpected errors?

• What tests have you written, and what gaps remain?

• What would you measure or change if usage increased significantly?

• What did you personally implement, what did you learn, and what can you demonstrate live?

Portfolio honesty rule: List this project only after you have built and tested it yourself. It is completely acceptable for the first version to be small. Do not claim AI, cloud, scale, or performance features you did not implement and cannot explain.

6. Behavioral Interview Question Bank

Use authentic experiences from coursework, team assignments, personal projects, employment, or other real responsibilities. Do not invent metrics, leadership scope, or achievements. STAR is Situation (brief context), Task (your responsibility), Action (your specific steps), Result (what changed and what you learned). Aim for 60–120 seconds, with most detail in Action.

1. Tell me about yourself.

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

2. Why software engineering? Why this role?

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

3. Why this company and what product would you like to work on?

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

4. Tell me about a challenging technical problem.

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

5. Tell me about a disagreement with a teammate.

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

6. Describe a mistake or failure and what you changed afterward.

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

7. Tell me about receiving difficult feedback.

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

8. Describe a time you learned something quickly.

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

9. Tell me about working under pressure.

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

10. Describe a time you took ownership.

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

11. How did you prioritize competing deadlines?

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

12. Tell me about a time you improved a process or design.

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

13. Tell me about a time you helped a teammate.

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

14. Describe a project decision you would make differently now.

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

15. What is a technical weakness you are actively improving?

Practice: write 4–6 bullet points using real details; state your own actions distinctly from the team's; name a concrete result or honest lesson; prepare one follow-up detail.

Company adaptation

Before each interview, review the job description and official company product/careers pages. Pick one real product area and connect your interests to the role. Avoid generic praise. Be ready to explain why the team's work, technical problem, and internship learning goals match your interests. Never assume a particular company uses a specific interview format until its current posting or recruiter confirms it.

7. Recruiting and F-1/CPT Checklist

Recruiting calendar (verify live openings)

Period	Action
Oct 14–31, 2026	Audit resume and GitHub; identify Summer 2027 roles; apply promptly to still-open roles; save the official job URL and date checked.
Nov 2026	Apply to open roles on a rolling basis; practice online assessments; ask UNT Career Center about career fairs and employer events.
Dec 2026	Continue applications and follow-ups; prepare for screens during winter break; do not assume postings reopen on a fixed date.
Jan 2027	Refresh resume/project evidence; review new postings weekly; run timed OAs and company-specific preparation based on confirmed process.
Feb 2027	Maintain weekly application/networking cadence; schedule mocks; follow up professionally after reasonable intervals.
Mar 2027	Focus on active pipelines, interviews, and remaining openings; prepare questions for interviewers and verify work-authorization requirements.
Apr 1–11, 2027	Prioritize active interview pipelines and late postings; review offer/authorization process with UNT before committing to any start date.

Important deadline note: Summer 2027 recruiting windows vary by company, team, role, and location. Do not treat the calendar above as a list of company deadlines. Check each live official job posting and record the date checked. One official Google Summer 2027 SWE intern posting stated rolling review and an anticipated application window ending September 25, 2026, with possible earlier closure; that particular posting's stated window has already passed by this plan's start date, so check for other active postings rather than assuming it remains open.

Application tracker columns

Company | role title | official job URL | location | date checked | deadline/window shown | status | last contact | next follow-up | OA/interview date | work-authorization wording | notes. Apply early to live postings because many are rolling, but never invent a deadline or claim an application is open without checking.

F-1 / CPT checklist — read before accepting or starting work

• Before assuming CPT is available, contact UNT International Student and Scholar Services (ISSS) and your academic advisor/DSO with the specific degree program, internship, course, dates and employer. Get individualized guidance in writing.

• UNT's current CPT page describes a significant regulatory update: federal guidance issued August 24, 2026 states CPT may only be approved when the practical experience is a required part of the curriculum and required for all students in the program to meet degree requirements. UNT says it is reviewing program requirements. This is an evolving area; confirm current interpretation directly with UNT ISSS.

• Do not begin employment, training, or internship activities before receiving the required CPT authorization and updated I-20 with the employer and authorized dates. An offer alone is not work authorization.

• Check the exact employer, work location, dates, hours, required course registration, and any academic approval. Authorization is specific; do not work before the authorized start date or beyond the end date.

• Do not assume unpaid work is automatically exempt. Ask UNT ISSS about the specific arrangement before performing work.

• Keep copies of the job posting, offer, advisor/ISSS approvals, course enrollment, and CPT I-20. Do not rely on advice from other universities or generic online summaries for UNT-specific eligibility.

Official sources

UNT International — Curricular Practical Training (CPT)

UNT Career Center — Internship Information

Google Careers — Software Engineering Intern, BS, Summer 2027 (example posting)

8. Daily Study Tracker and Final Readiness

Copy this log into a spreadsheet, notes app, or paper journal for every day.

Field	Record
Date / day number	
Planned topic(s)	
Actual focused minutes	
New problems attempted	
Solved independently	
Needed hints	
Needed full solution	
Topics/problems revised	
Mistakes discovered	
Confidence (1–5)	
Mastery status	Not Tried / Tried / Developing / Strong / Mastered
Next review dates	R1 / R3 / R7 / R14 / R30 / R60 from original learning date
Tomorrow's top priority	

Final readiness checklist (Day 180)

■ I can solve representative easy and medium problems independently across arrays/hashing, two pointers, windows, binary search, linked lists, stacks, trees, heaps, backtracking, graphs and DP.

■ I can explain a brute-force baseline, derive an optimized approach, prove the core invariant, state time/space complexity, and test edge cases.

■ I use Java collections correctly, understand equals/hashCode, nulls, integer overflow, recursion stack, and PriorityQueue ordering.

■ I can explain OOP/SOLID, processes/threads, synchronization/deadlocks, SQL/indexes/transactions, HTTP/DNS/TCP, and basic backend reliability.

■ I can design a simple service, define APIs/data models, and discuss caching, replication, consistency, load balancing, queues, and observability at internship level.

■ My backend project runs from documented instructions, has meaningful tests, a clear README, architecture notes, and honest limitations.

■ I have 6–8 authentic STAR stories, a concise resume walkthrough, and a practiced project deep dive.

■ I maintain an accurate recruiting tracker and verify each job's live status, deadlines, and work-authorization requirements.

■ I have contacted UNT ISSS/DSO about my individual F-1/CPT situation and will not begin employment without required authorization.

Pass criteria and remediation

A reasonable final benchmark is at least 4/5 in independent DSA solving, explanation, Java correctness, CS/design fundamentals, project deep dive, and behavioral clarity, with no unresolved critical issue. If a category is below 4, spend the next two weeks on 60% targeted repair, 25% mixed timed practice, and 15% project/recruiting maintenance. A score is a planning signal, not a guarantee of an offer.

9. Problem Index and References

Problem bank used across the daily schedule

Java foundations

• Valid Anagram — Easy; frequency counting.

• Contains Duplicate — Easy; HashSet membership.

• Two Sum — Easy; HashMap complement.

• Best Time to Buy and Sell Stock — Easy; one-pass invariant.

• Ransom Note — Easy; frequency counting.

Arrays and hashing

• Group Anagrams — Medium; canonical key / hashing.

• Top K Frequent Elements — Medium; frequency map + heap/buckets.

• Product of Array Except Self — Medium; prefix/suffix products.

• Longest Consecutive Sequence — Medium; HashSet sequence starts.

• Encode and Decode Strings — Medium; serialization boundaries.

Two pointers and windows

• Valid Palindrome — Easy; two pointers.

• 3Sum — Medium; sort + two pointers.

• Container With Most Water — Medium; greedy two pointers.

• Longest Substring Without Repeating Characters — Medium; variable sliding window.

• Minimum Window Substring — Hard; window counts.

Binary search and sorting

• Binary Search — Easy; search invariant.

• Search in Rotated Sorted Array — Medium; modified binary search.

• Find Minimum in Rotated Sorted Array — Medium; partition invariant.

• Koko Eating Bananas — Medium; binary search on answer.

• Merge Intervals — Medium; sort + sweep.

Linked lists, stacks, queues

• Reverse Linked List — Easy; pointer rewiring.

• Linked List Cycle — Easy; fast/slow pointers.

• Merge Two Sorted Lists — Easy; dummy node.

• Valid Parentheses — Easy; stack.

• Daily Temperatures — Medium; monotonic stack.

Trees

• Maximum Depth of Binary Tree — Easy; DFS recursion.

• Invert Binary Tree — Easy; recursive tree transformation.

• Binary Tree Level Order Traversal — Medium; BFS queue.

• Validate Binary Search Tree — Medium; bounds / invariant.

• Lowest Common Ancestor of a Binary Tree — Medium; postorder reasoning.

• Binary Tree Maximum Path Sum — Hard; tree DP.

Backtracking, heaps, greedy

- Subsets — Medium; backtracking decision tree.
- Combination Sum — Medium; backtracking + pruning.
- Word Search — Medium; grid DFS + backtracking.
- Kth Largest Element in an Array — Medium; heap / quickselect.
- Task Scheduler — Medium; greedy counting.
- Implement Trie (Prefix Tree) — Medium; prefix tree.

Graphs

- Number of Islands — Medium; grid DFS/BFS.
- Clone Graph — Medium; DFS/BFS + map.
- Course Schedule — Medium; cycle detection / topological sort.
- Rotting Oranges — Medium; multi-source BFS.
- Redundant Connection — Medium; Union-Find.
- Network Delay Time — Medium; Dijkstra.
- Min Cost to Connect All Points — Medium; minimum spanning tree.

Dynamic programming

- Climbing Stairs — Easy; 1D DP.
- House Robber — Medium; take/skip recurrence.
- Coin Change — Medium; unbounded knapsack.
- Longest Increasing Subsequence — Medium; DP / binary-search optimization.
- Longest Common Subsequence — Medium; 2D DP.
- Unique Paths — Medium; grid DP.
- Word Break — Medium; prefix DP.
- Edit Distance — Medium; 2D DP transitions.

Bit manipulation and mixed

- Single Number — Easy; XOR properties.
- Counting Bits — Easy; bit DP.
- Find the Duplicate Number — Medium; cycle detection / value search.
- Search a 2D Matrix — Medium; binary search.
- Median of Two Sorted Arrays — Hard; partition binary search.

Interview communication checklist

- Clarify input, output, constraints, duplicates, and edge cases.
- Describe a simple baseline before optimizing.
- State the invariant or reason the pattern applies.
- Walk through one small example before coding.
- Write readable Java; name variables by purpose; avoid unnecessary cleverness.
- Test normal, boundary, empty, duplicate, and adversarial cases as relevant.
- State time and space complexity precisely, including auxiliary space.
- Summarize the result and mention a trade-off or follow-up optimization.

Selected official reference links

- UNT CPT page
- UNT Career Center internship information

- Google Summer 2027 SWE intern posting example
- LeetCode
- Spring Boot documentation
- JUnit 5 User Guide
- PostgreSQL documentation

Quality-control note: The program contains 180 dated entries, each with a 180-minute schedule, a concrete anchor exercise, a deliverable/exit criterion, and calculated anchor review dates. The six review intervals are derived from each anchor's original day. Company deadlines are not fabricated; official postings must be rechecked because openings and windows change.